

#LodzfeelsIT

Lodz IT Ecosystem Report 2022

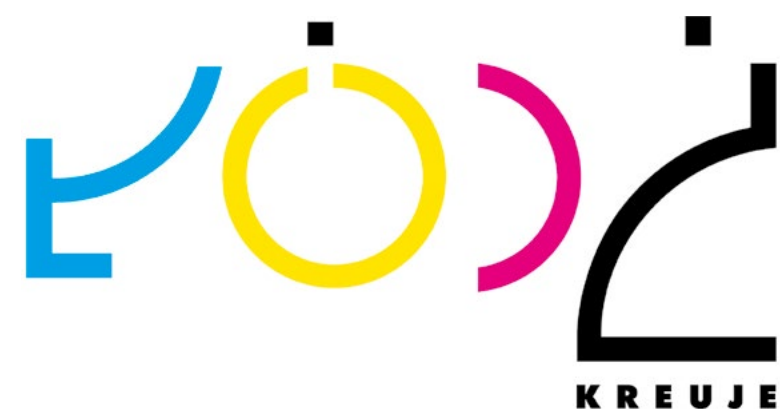
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POLAND
CLUSTER

ŁÓDŹ
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Each year, Lodz records dynamic development, thanks to which it is an attractive place for business on the map of Poland and Europe. The City of Lodz focuses on technology development and creating a favorable climate for the IT sector. It is no longer a place where only costs are optimized. We have become a city that can provide quality and specialized staff. Lodz has a lot to offer in this area - both high and middle level managers and specialists. The City creates the best conditions for life and career development and supports activities that enable it.

The dynamics that characterize modern branches of the industry require constant monitoring and understanding of its environment. The development of the report was one of the priority tasks of the City. The report creation was possible thanks to the commitment of its exceptional Partners - Grafton Recruitment, CBRE, and the ICT Central Poland Cluster. They have provided us with key figures and insights.

Thanks to such extensive cooperation, the report paints a broad picture of the IT industry in Lodz - a diagnosis and signpost for the innovation ecosystem development. We hope this document will be a crucial source of information on the Lodz market of new technologies and its development trends soon.

Adam Pustelnik

Deputy Mayor, City of Lodz Office



IT and technologies are taking the labor market in Lodz by storm. Lodz is a city whose development dynamics are high in every respect. Recent years have changed the face of the City – not only in terms of subsequent revitalization and infrastructure projects – but also had an equally strong impact on the business and, consequently, the labor market.

Recent years have been an enormous leap toward IT and technology for the City. It would not be possible without the close cooperation of the world of science and business, as well as without the exceptionally proactive attitude presented by Business Development and International Relations Bureau – Lodz’s catalyst for business activities.

It was also accelerated by the pandemic, stimulating in employers a change in budgetary issues perception and talent retention approach. In addition, it has forced a new approach to designing recruitment activities. Although the globalization of the IT industry took place before, also in Lodz, the pandemic gave rise to the need to enter new markets, diversify business risk, and analyze the prospects of job candidates. At the same time, it facilitated making business decisions in the IT industry, which means that the technology market in Lodz is on an upward wave. There is no indication that it will stop.

At Grafton Recruitment, we conduct about 1,600 recruitment processes for our clients annually. We observe the great interest of candidates in job offers in Lodz. Interestingly, people planning to relocate from abroad also indicate Lodz as a destination where they would like to live and work.

We translated our experiences and observations into our contribution to the Report prepared by the City of Lodz Office’s Business Development and International Relations Bureau. In the Report, you will find information on the Lodz IT market in a nutshell, including current trends and an assessment of the prospects for the IT sector development in Lodz in the next 12 months.

Mariusz Lemparty

*Branch Manager Lodz,
Grafton Recruitment*





From the real estate market perspective, Lodz meets all the conditions necessary for dynamic business development. Moreover, the office spaces (with modern standards and in attractive locations) are a kind of offer that appeals to foreign investors and leading business organizations from Warsaw and other Polish cities.

Excellent quality to rental costs ratio significantly distinguishes Lodz on the business map of Europe and Poland. It gives the local estate market a solid competitive advantage.

Most importantly, however, thanks to the local labor market and the involvement of the City of Lodz Office, Lodz has become one of the most attractive places to work and live. It is a prerequisite for the further dynamic development of the city. The report partners - The Grafton Recruitment and the ICT Cluster - will tell you more about it.

Lodz has become one of the most attractive places to work and live. It is a prerequisite for the further dynamic development of the city.

Jan Banasikowski

*Associate Director
CBRE | Advisory & Transaction
Services*



The ICT Central Poland Cluster was established to integrate the Lodz IT environment and strengthen cooperation between enterprises and universities. Although cluster participants are often market competitors, we have worked together for Lodz and its region development. Our goal is to ensure high-quality human resources for the IT industry in Lodz and to promote the potential of companies, universities, and the city of Lodz itself.

We have a significant and recognizable voice in the discussion about the industry's future in the region. As a cluster, we are a source of initiatives supporting innovation and cooperation of economic, academic, and scientific communities. We also provide substantive and opinion-forming support for projects submitted by other entities participating in economic life.

One of the main goals of the cluster is to improve education in information technology and computer science at Lodz universities for the needs of a modern economy and high technology industry. Collaboration between universities and employers is essential, especially, when technology changes as rapidly as in the IT and telecommunications industry. All parties appreciate these activities in education, and the availability of highly qualified staff is one of the critical factors for the further development of innovative ventures in Lodz. **Thanks to the cooperation between universities, employers, and the authorities of the city of Lodz, we create the best possible living conditions and career development for IT specialists.**

Prof. Paweł Strumiłło

Chairman of the ICT Central Poland Cluster



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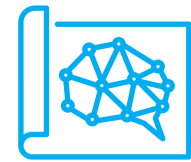
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Executive summary

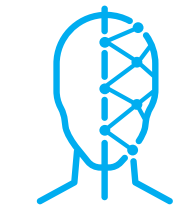


29 500+ IT specialists in Lodz Agglomeration

including DevOps, Cloud and Big Data Engineers, Machine Learning and AI specialists, experienced Frontend & Backend Developers, able Testers and System Architects



Over 200 IT companies in Lodz IT sector



98% IT companies

plans to increase the employment of IT specialists in the next 12 months



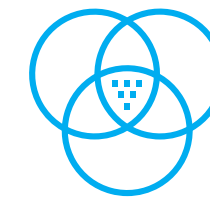
1100 IT students

graduate from universities in Lodz every year



+82%

demand* for IT specialists compared to Q4 2019 in Lodz



Almost 55%

lease contracts belong to the IT sector



40

members of the ICT Central Poland Cluster



611 600 m2

of modern office space

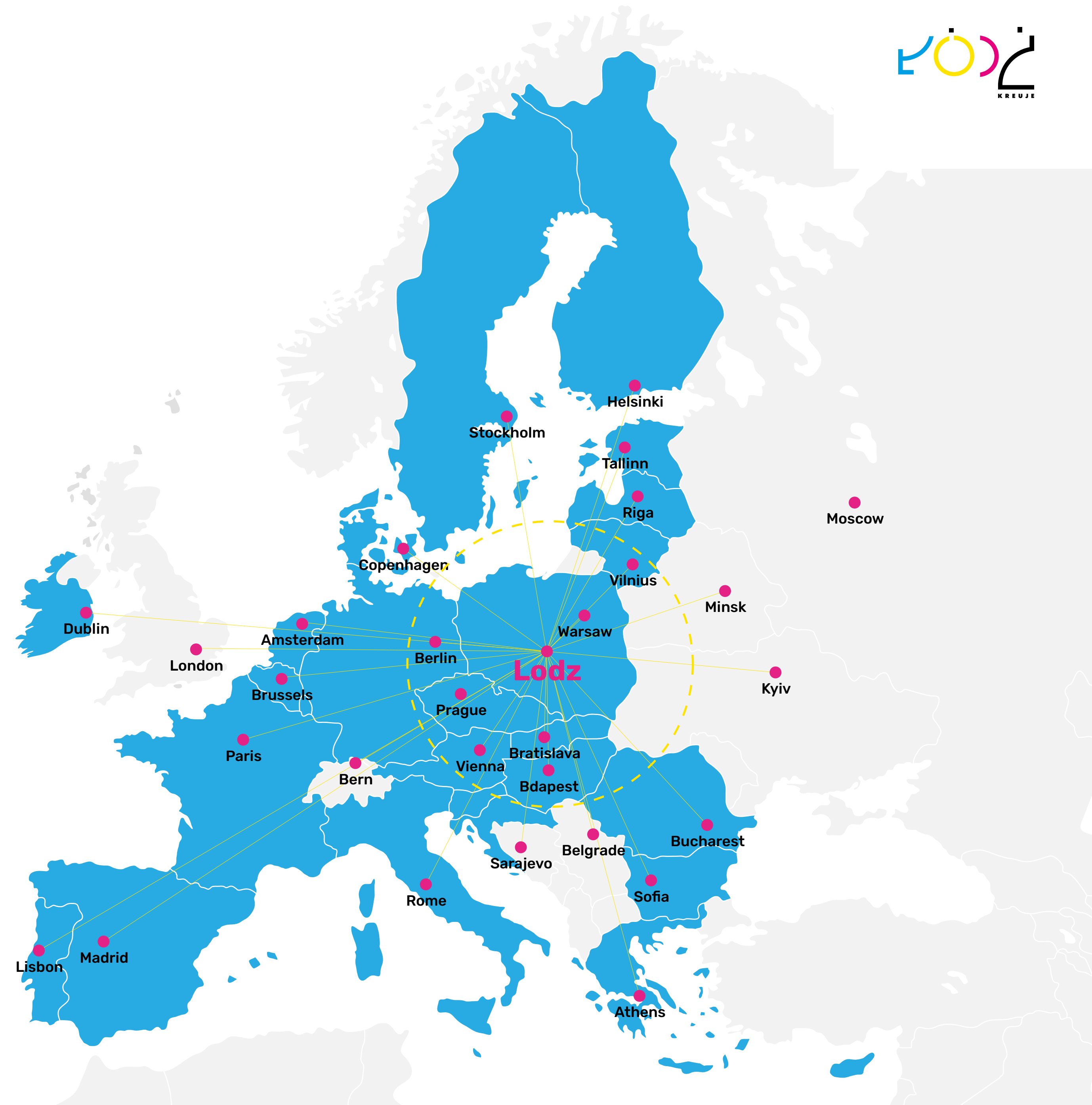
**It is ahead of the benchmark for the whole of Poland by 43 percentage points*

Lodz

Key information



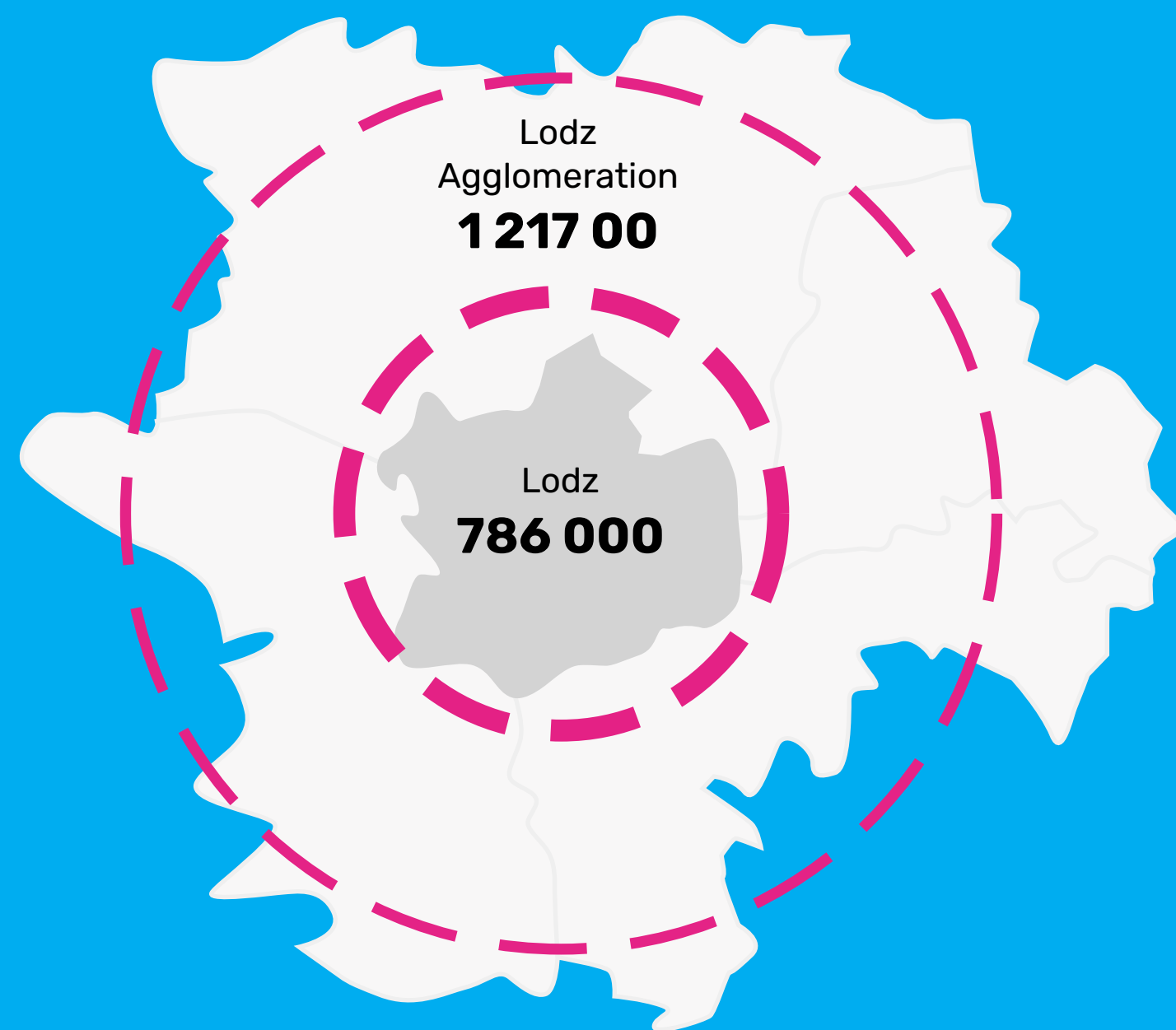
Centrally located
& easily accessible*



*INTERNATIONAL AIRPORTS: Reymont Airport 15min.
from Lodz city center Chopin Airport 90min.

The third biggest city in Poland

Population according to the information for 2021 provided by the Central Statistical Office and the data collected by the Union of Polish Metropolises in the report: *Urban hospitality: unprecedented growth, challenges, and opportunities*



CEE's most business-friendly city

Emerging Europe 2021 business-friendly city perception ranking based on the FDI Experts, site selection advisers and location analysts

1st place in the overall ranking



And recognition in the following categories:

- Business climate - 1st place
- Local authority support - 1st place
- Infrastructure and connectivity - 3rd
- Economic potential - 4th
- Brand - 4th
- Talent pool - 6th





FDI
FREE ZONES
OF THE YEAR
2020



LODZ
SPECIAL
ECONOMIC
ZONE



FREE ZONE GLOBALLY



EUROPE



GLOBALLY FOR SME



EUROPE FOR SME



5G
READINESS



CONNECTIVITY



FUTURE
MOBILITY



FREEZONE
4.0

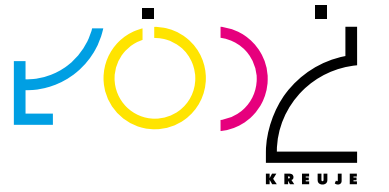


INDUSTRY
4.0

The best economic zone in Europe

The best use of the European funds in Poland
The highest level of public aid in Poland*

** The amount of public aid granted to IT, BPO&SSC
companies in the form of CIT exemption*



Lodz

IT ecosystem



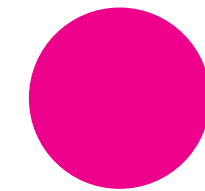
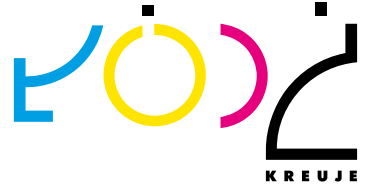
Over **200 IT companies** trusted Lodz and keep developing their businesses here

Current IT talent pool



IT specialists

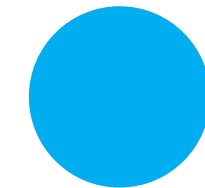
Lodz talent pool in numbers



LinkedIn Statistics

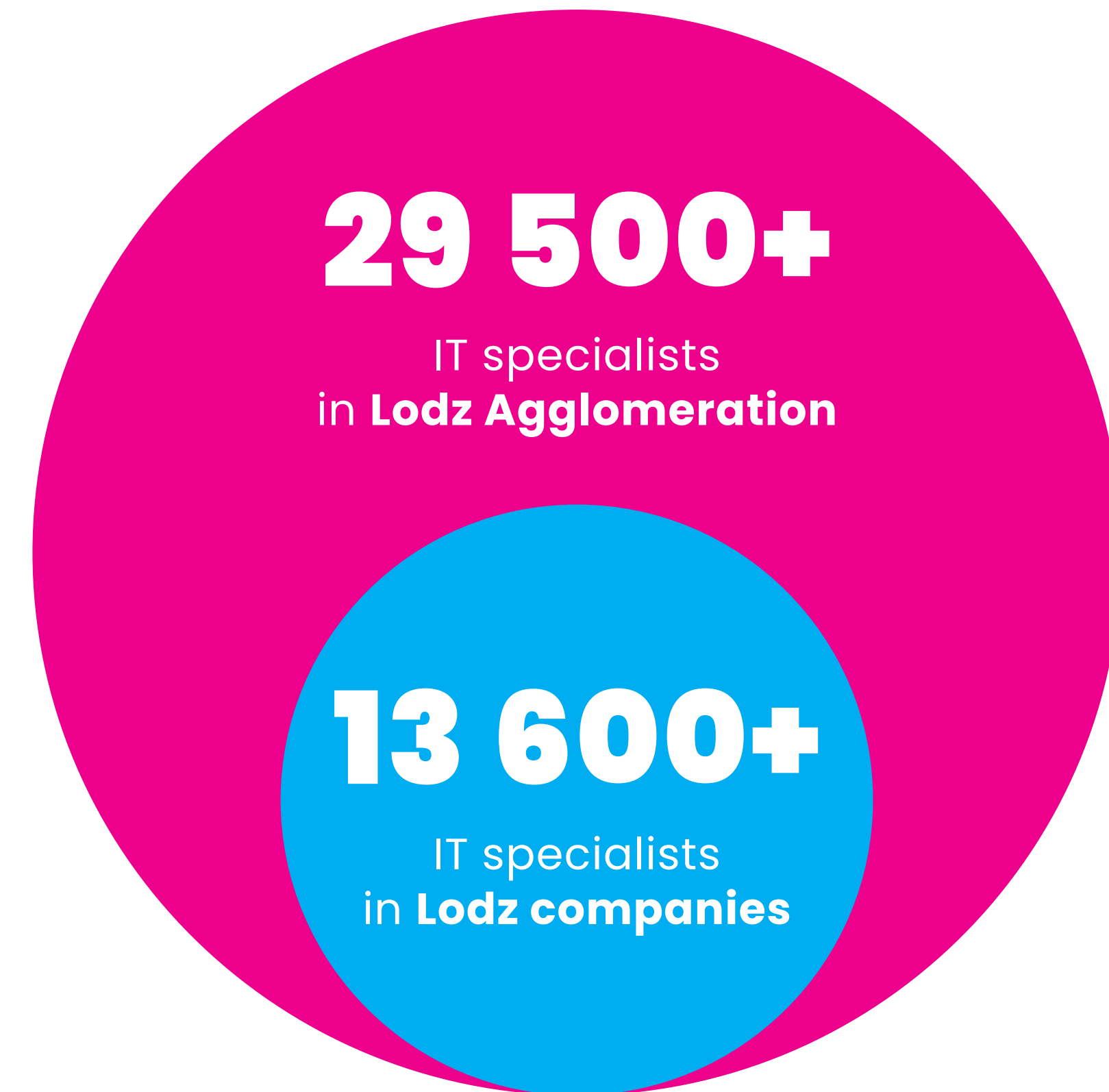
Location – Łódź Agglomeration

Skills and position that indicates that they are IT specialists



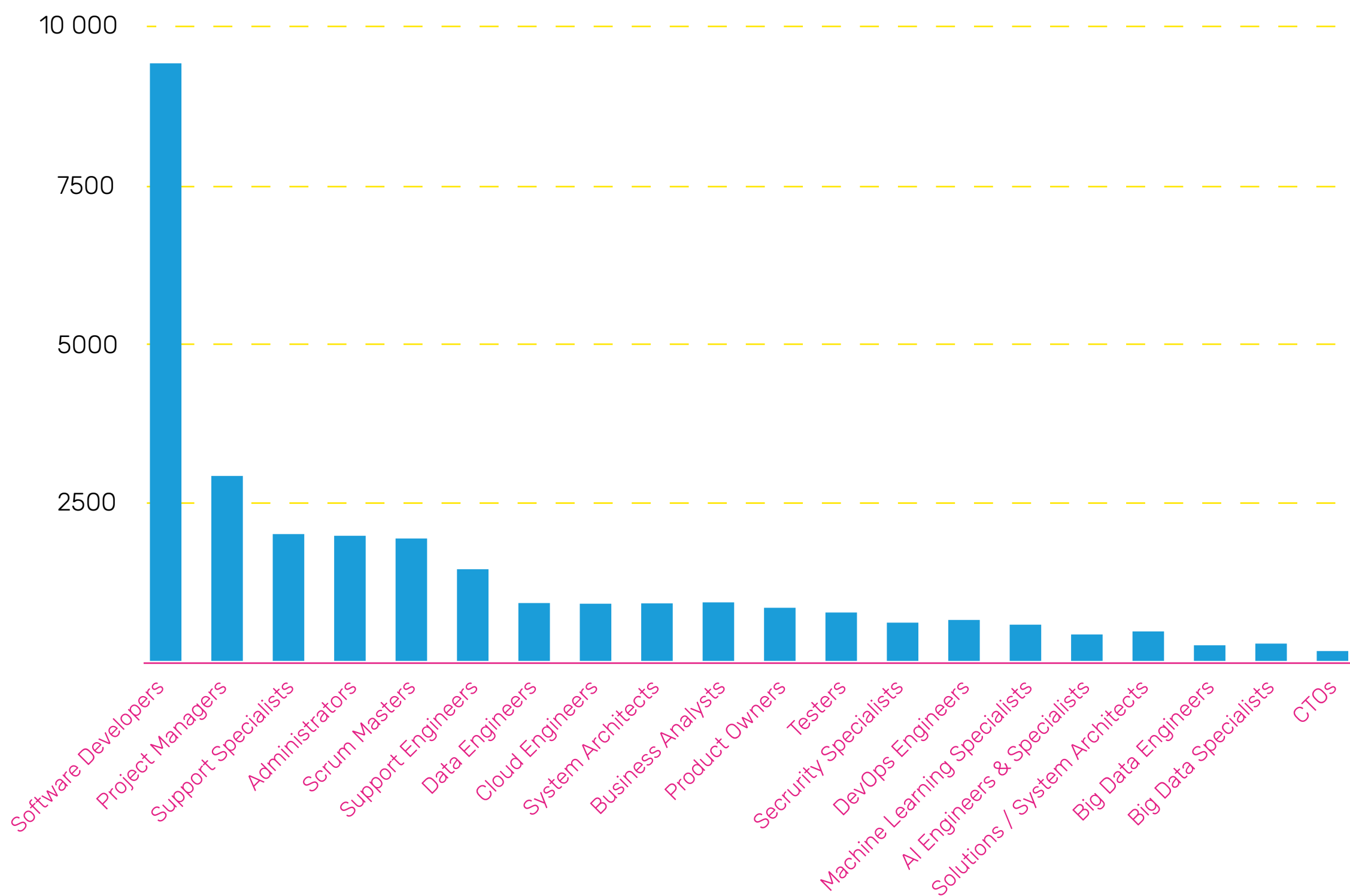
Survey & Estimate

Done among Lodz IT companies and SSC companies with a strong IT department*



29 500 IT specialists in Lodz Agglomeration

Number of IT specialists in Lodz Agglomeration by JOB TITLE*
according to LinkedIn Sales Navigator data [current job title only]



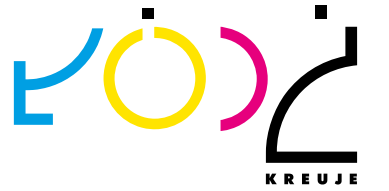
**LinkedIn Statistics based on Sales Navigator Tool:*
 - Location Lodz Agglomeration
 - Skills and positions that indicating being an IT specialist



Continuous IT talent pool development



IT education



Nearly **5000 IT sciences students**



Over **1100 skilled IT graduates every year**

Technology-friendly municipal authorities play a vital role in IT sector fast development, as well as the openness of local universities to cooperation with business.

Higher education institutions are constantly adapting their offer to the needs of business. Universities created programs like Modelling and Data Science, Artificial Intelligence and Machine Learning, or Human-Computer Interaction.

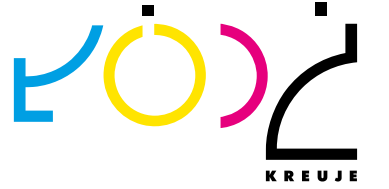
19 universities

30 R+D institutions

5 universities offering education in the **IT field**



University of Lodz cooperation with business



Innovative programs built together
with the sector leaders

Business Processes Automation

BA degree

The initiating partner is Nordea, while the group of partners currently includes: Fujitsu, Digital Workforce, Digital Teammates, DXC Technology, B/S/H, Mindbox. Representatives of these companies conduct classes for students. The program is offered at Faculty of Management and supervised by: the Artificial Intelligence and Cyber Communication Research Center and the Smart Technologies Center.



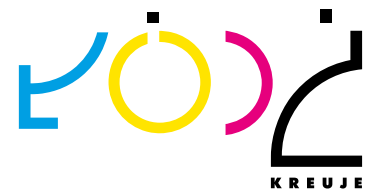
FinTech

POST GRADUATE degree

Seven FinTech companies signed a preliminary cooperation agreement with the University of Lodz under the patronage of Cashless Poland. The purpose is to build a fully integrated ecosystem of FinTech companies that bring the unique expertise of their business experience to the study program.



Idea for a startup acceleration program

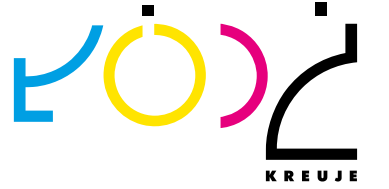


For innovative, science, technology, and creative projects at the different stages of development

- 13 editions
- 40 projects presented to venture capital fund
- 88 projects covered by the mentoring program
- 350+ founded companies
- 800+ projects covered by training&advisory support
- 1550 submitted projects



Youth in Lodz scholarship programs



Mayor's scholarship program

Funded by the Mayor of the City of Lodz Office

For whom?

For the best high school graduates who decided to study in Lodz.
All scholarship recipients are subject contests winners and finalists.

Employers' scholarship program

Funded by Lodz companies

For whom?

For the best students of Lodz universities, chosen
according to the needs and preferences founder companies

Purpose

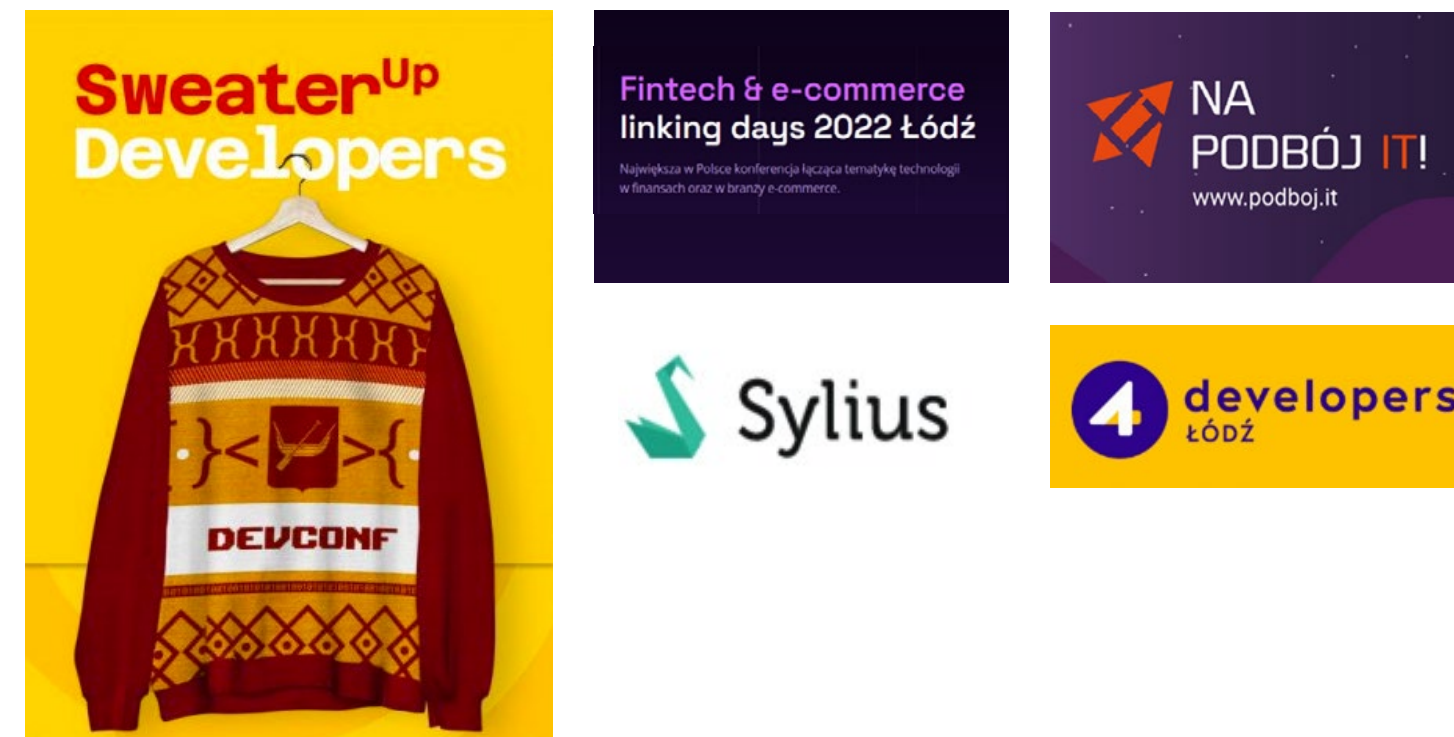
To attract and keep the most talented people in Lodz. People who will enjoy studying, working, and living here.

Over 30 meetups
& various hackathons



Conferences for the IT sector

Events co-created with the City of Lodz Office



International events with a strong Lodz IT Ecosystem presence



The City of Lodz - the only Polish city to have its official stand during the WebSummit Conference.



LODZ

Lifestyle



City with character

Lodz is a large city without inherent metropolitan tension. It is easy to get from place to place. Employers and employees appreciate the compact nature of Lodz, which provides them with the right work-life balance. **Unique post-industrial urban space, the famous red brick, high-quality coworking spaces, and remarkable office spaces stimulate creativity and promote integration.**

The cost of living – competitive compared to other large cities – makes Lodz an attractive alternative for young workers and entrepreneurs looking for work comfort and high quality of life at an affordable price.

Lodz is also strongly associated with openness. Citizens of Lodz quickly shorten the distance. The traditions of the four cultures – which built the city – make everyone feel instantly at home. No matter where you come from – if you have an idea and energy to act, you will quickly become part of the Lodz community.



City with character



Place to live, place to grow

The unique atmosphere of Lodz accompanies me in my private and professional life. Being a Lodz native, I consciously chose this city to create and develop my company. Today I am proud that our team – of nearly 120 people, appreciated in Poland and abroad – is proof that it is possible to create an international company in Lodz. It is precisely a magnet that makes more people migrate to Lodz. In addition, the exchange of knowledge and experience among Lodz IT companies is an invaluable source of inspiration for continuous further growth.

Maciej Gałęcki

CEO, Bluerank

Lodz is the cradle of intercultural dialogue, connecting business, science, sport, and art. It is a city based on intergenerational creativity. Building bridges between different needs based on respect, readiness for change, and curiosity are part of the value of the inhabitants. Lodz academic centres – solidly positioned among the rankings of Polish universities and tied with the business and start-up ecosystem – are a continuous source of innovation. Those centres stimulate the growth of entrepreneurship and are a driving force for sustainable city development. Lodz draws its strength from openness to diversity, geographical position, and the continuous process of improving the quality of life in the city.

Przemysław Zakrzewski

Head of Corporate Technology Center, ABB



In Lodz IT companies feel they are equal partners in a dialogue with the city. The City of Lodz Office appreciates our opinion and consults with us.

In my opinion, Lodz - a city with an industrial past - fits perfectly on the map of the Polish IT market today. The support from the City of Lodz Office (that investors and companies already operating here can count on) plays a significant role. In Lodz, IT companies (and those from other industries) feel they are equal partners in dialogue with the city. The City of Lodz Office appreciates our opinion and consults with us. We share the point of view of investors in our industry and help Lodz to attract new companies faster. It is good news because greater competition means more emphasis on improving the quality of services and products and introducing innovations in this area - something the IT industry has in its blood. The effects of the City of Lodz Office operation are visible to the naked eye - only in 2021 was it possible to bring a dozen investors to the city, including several companies operating in the IT market.

City of Talent

Lodz is also a center of talent and a broad academic base that sees value in cooperation with business. I know from experience that universities want their graduates' competencies to meet the labor market requirements. For example, I can mention an initiative that took place for the first time more than ten years ago: a test of IT students organized in cooperation with universities in Lodz. As a task co-author, I have watched how young people deal with questions created by practitioners. Although the first edition brought excellent results, the results achieved by students improved from year to year. From the perspective of a company operating in the IT

market, we can only be happy - this means that the programs implemented in the studies are practice-oriented. It is no coincidence that a team consisting mainly of Lodz GFT employees, also graduates of local universities, created the world's first utterly virtual bank - Mox. As the above examples show, cooperation between businesses and universities can only bring positive results. It is worth going in this direction and finding new levels of understanding - there are undoubtedly many more.



Good conditions for stable development

A significant advantage of Lodz is the broad base of employees with junior competencies – young graduates or people ready to change industries. Whoever works in the IT industry knows perfectly well that it is struggling with frequent rotations. So we are all the more pleased that most of our employees in Lodz have five and ten years of work experience. It is a result of our culture and the significant number of development and certification programs we run at GFT. We believe that by providing our employees with good conditions from the beginning, we can build a team that will stay with us for many years.

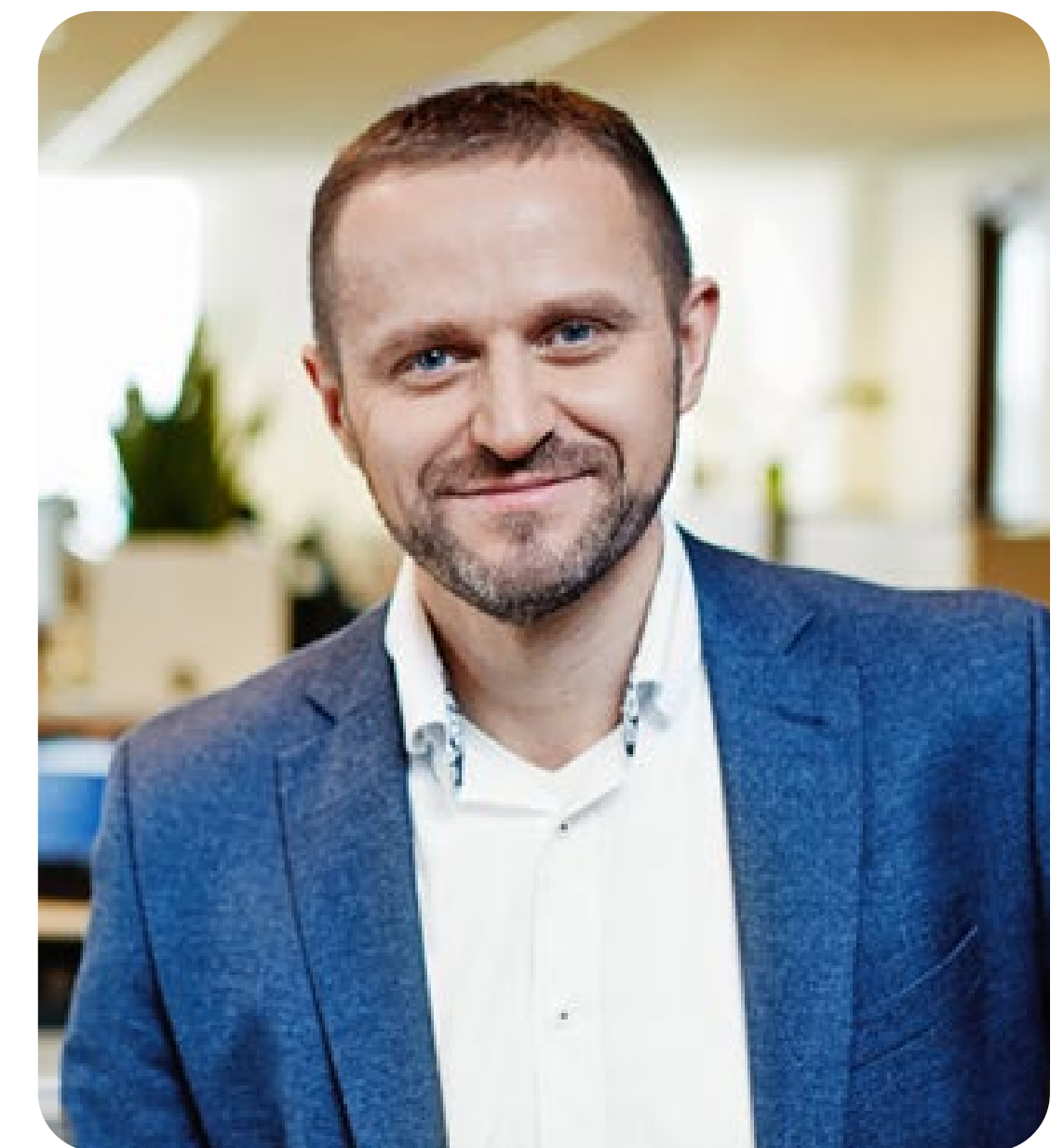
Offices where you want to meet

Whoever visits Lodz knows that the city has experienced a renovation and development boom in recent years. The base of high-quality office space is growing, also used by GFT. We are based in Lodz at Brama Miasta. The office perfectly suits our needs, and we used the time during the pandemic to introduce additional amenities and prepare it for the new reality. We know that the hybrid mode will stay with us for many years, which is why we have remodeled the office with the changing needs of our employees in mind. Our office is where you want, and you don't have to come. We treat our office as a base and a place where we can always come to work, talk or meet the GFT team professionally or privately. We have conference rooms, relaxation rooms, and a hot-board system for those who do not have the conditions for remote work at home. The location of both the office and the city is also a significant advantage. Its central location, the railway station in the business part of Lodz, and the ability to quickly move around the city make it highly attractive. If someone would like to stay in Lodz for a longer time, he will soon find out that the costs of living and buying an apartment are much lower here than in the capital.

City and business hand in hand

Our internal company survey shows that as much as 86% of our employees are satisfied with their work at GFT. Employees appreciate the conditions offered by GFT. Yet friendly and efficient companies do not arise in a vacuum. It combines good infrastructure, location, development conditions, and cooperation with the city authorities. It is precisely how the Lodz IT ecosystem works!

Marek Bartosik
COO, GFT



A significant part of my professional life is related to Lodz. I have had the opportunity to build and transform organizations of various sizes – from start-ups to four-digit giants. There were many good stories and important lessons learned, but the crucial thing is that I keep returning to Lodz to carry out new projects.

IT job market in Lodz – although not as large as in Warsaw or Kraków – is more focused on long-term development than opportunistic actions. Lodz has a mature IT ecosystem that efficiently connects employers and universities. An ecosystem that influences the shape of the future IT talent pool. Quantifiable and sustainable growth at a highly predictable pace allows you to plan and achieve goals with high predictability. In addition, the central location facilitates the swift maturing of IT companies. The remote-hybrid work enables the combining of talents coming from many Polish cities.

You want to be in Lodz and you want to build new organizations right here! The Lodz job market is much bigger than the information you can see in the brochures. On the other hand, a nice boost to the number of international connections from Lodz airport would be a much-appreciated development. Though, if you need to fly to cities other than London, Dublin, Milan, Brussels (or one of the summer destinations), reachable within 90 minutes by car Chopin Airport comes in handy.

When deciding to locate the First Derivative office in Poland, the many features of Lodz IT ecosystem, including the quality of its talent pool helped us to make the right decision. We are mainly building an IT company. That will appeal to all those who enjoy shaping reality around modern technologies. In addition, we are establishing department providing banking business services such as KYC or AML.

We are looking for experienced specialists. In parallel, we also focus strongly on creating competencies from scratch. Lodz universities are a great bargaining chip for the city. I am sure that with the help

of organizations such as the City of Lodz Office's Youth in Lodz programs and direct work with universities, we will launch many early career-building programs. It is a great starting point for both technical and non-technical graduates. The sky is the limit!

Lodz has a mature IT ecosystem that efficiently connects employers and universities. An ecosystem that influences the shape of the future IT talent pool.

Szymon Kaczmarczyk
Managing Director, Head of Engineering Services, First Derivative

**FIRST
DERIV
ATIVE**



Lodz from the perspective of **#LODZfeelsIT** Report's Content Partners





Research: Lodz from the perspective of employers



To get to know the specificity of the IT market in Lodz, together with the City's Business Development and International Relations Bureau, we surveyed Lodz enterprises.

We asked about the following:

- Employment (number of employees working as IT specialists)
- Technologies used
- Recruitment plans for the next 12 months
- Evaluation of the effectiveness of remote work

We divided the obtained responses into three groups, distinguished based on the size of employment, i.e.:

- 0-50 employees (25 entities)
- 50-250 employees (19 entities)
- 250+ employees (18 entities)

The information collected in the questionnaires was used by us, among others, to identify technologies popular among companies (expressed in general categories), to learn about their recruitment plans for the next year, and to look at their subjective assessment of the effectiveness of remote work.



Research: companies that participated in the survey



Lodz IT sector recruitment potential

In terms of the diversity of processes, Lodz already showed a few years ago that you could look for specialists in many areas of finance or interdisciplinary roles that require competence in the field of, e.g., analytics and supply chain. The last time Lodz went through a significant development boom in the BPO/SSC industry context. New investors saw the great market potential, location attractiveness, and the need for business diversification. It has resulted in choosing Lodz as a perfect place to develop their business and projects.

The lack of specialists' availability to implement subsequent projects affects the entire IT industry. Still, the multitude of investments made recently in the Lodz IT market shows that companies perceive this market's advantage and the city's attractiveness in terms of the candidate's choice as a place to live. This recent history of successful investments of such brands as Godel, iTechArt, Ericsson, ZF, Infopulse, and Global Logic proves that.

Ericsson, already present in Lodz, has invested in a project related to 5G and ZF, developing new laboratories and, thus, new jobs in the IT industry. Before the outbreak of the war in Ukraine, Infopulse increased employment by relocating workers. Global Logic – an IT service provider for the industry, chose Lodz as the direction of its expansion.

An additional and undeniable advantage of Lodz is the changes taking place in it. The revitalization of the city, the transformation of the centre, and new business centres (like Monopolis), which collect praise and awards around the world, attract new residents and encourage them to live and work in the centre of Poland, next to the capital of our country.

It is also worth looking through the eyes of a potential investor at the level of saturation of the local market and business scalability for the implementation of future projects. Furthermore, as the IT ecosystem is a strategically important sector for the city, we can assume that Lodz will maintain the dynamics of industry development in the coming years, which means an increase in employment and the number of new companies established in Lodz. Therefore further growth of the IT services market is a given.

The study of companies that we conducted together with the Business Development and International Relations Bureau at City of Lodz Office shows that the IT competencies market in Lodz is more extensive than it might seem after the analysis of the available data. Most importantly, however, it is diversified in terms of competencies, which proves its maturity and continuous development.

Therefore, the employment plan, based on the competencies offered by Lodz, combined with the popularity of remote work and the relocation of specialists coming to the city, is a powerful argument for investing in new IT projects right here.

Mariusz Lemparty

Branch Manager, Grafton Recruitment



Recruitment plans of Lodz IT companies

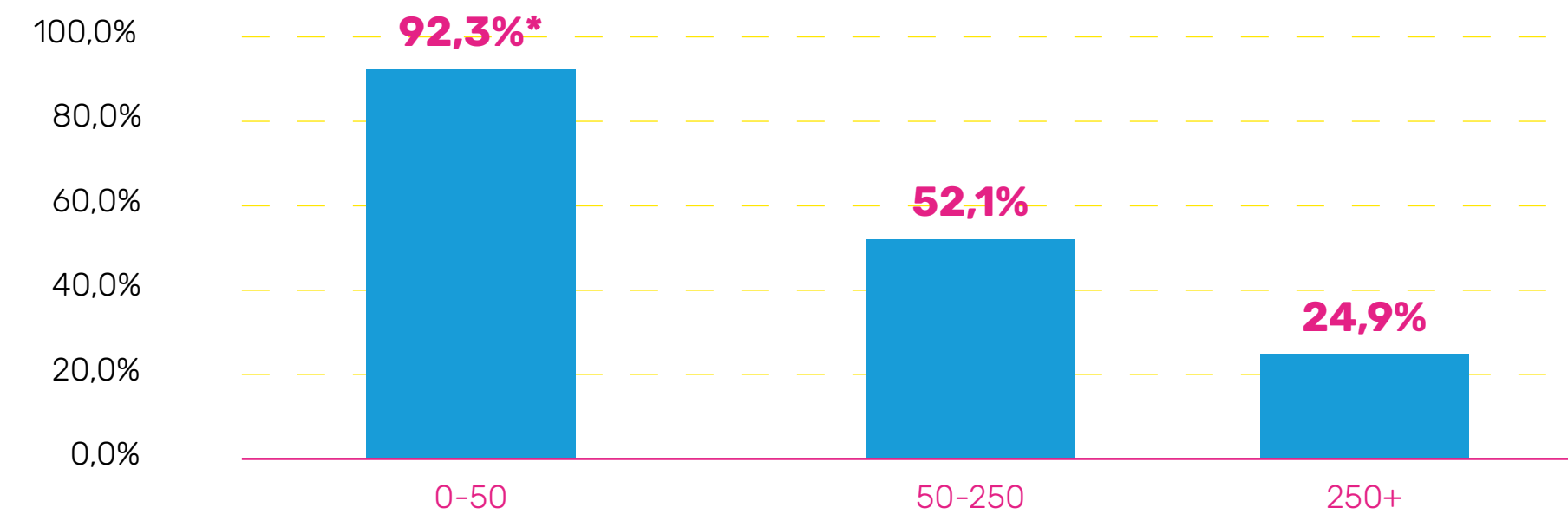
Our research confirmed the enormous recruitment potential of Lodz in the IT industry.

We asked if companies intend to expand their team with technical employees/IT specialists in the next 12 months. Sixty-one surveyed companies answered the question about their recruitment plans.

As many as 98% of the respondents answered this question in the affirmative: most employers in Lodz intend to increase employment and recruit specialists from the IT industry within a year.

Enterprises from Lodz assume that within one year, they will recruit 2,716 specialists for IT-related positions. Currently, the surveyed companies as specialists employ 8,852 people, so the declared annual growth is 30.7%.

A planned increase in the employment of specialists in the next 12 months (in%)



The chart shows the increase in the number of IT specialists declared by the respondents in the next 12 months, broken down into three groups of companies, ranked in terms of employment size.

* Data after rejecting the two highest and two lowest results
- due to the significant influence of companies that have just entered the Lodz market.

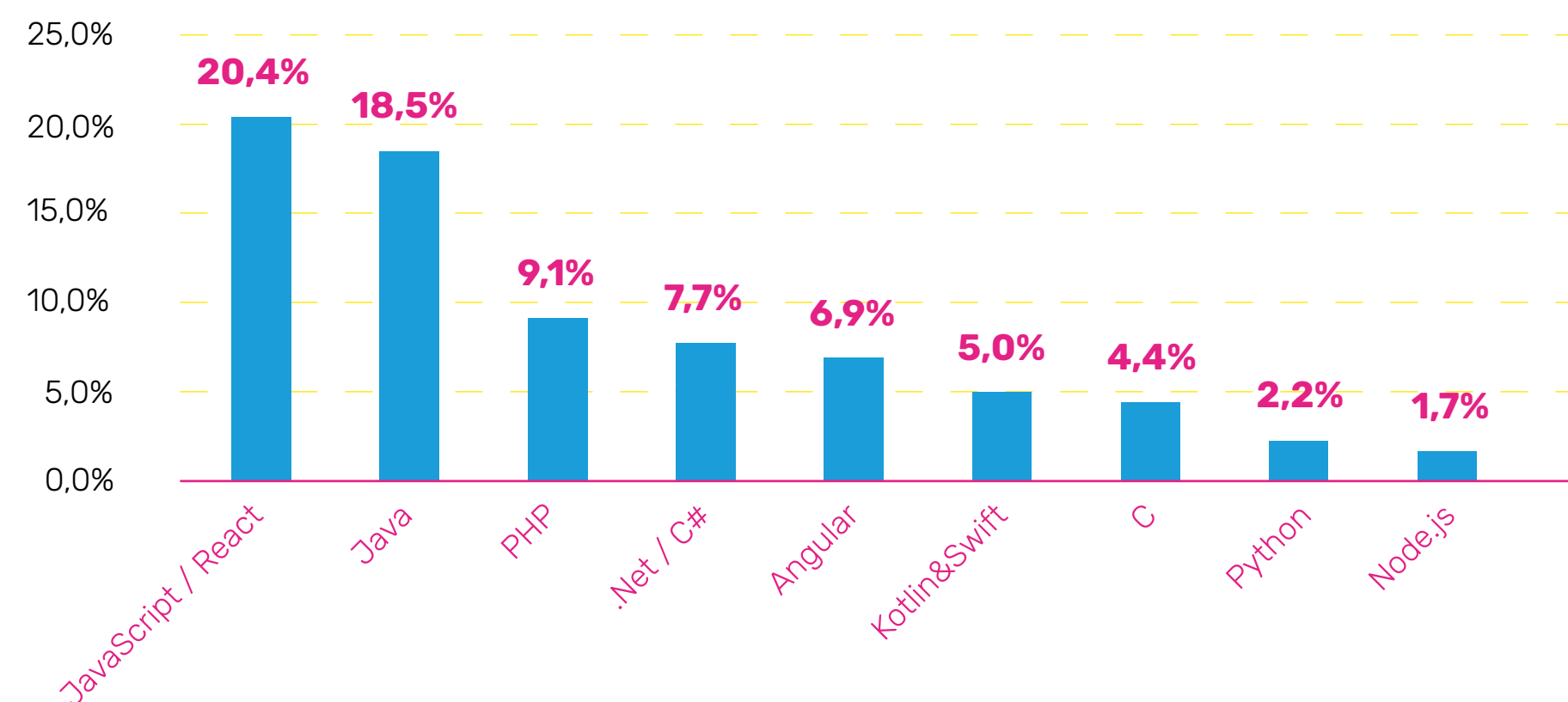


Popular technologies

In the other part of the survey of Lodz companies, we asked the respondents to identify the four most important technologies used in their enterprises.

Sixty-one respondents answered this question. We divided their answers into three groups based on the number of the company's employees. After assigning the data to the main categories, we noticed the following dependencies in each of the three groups:

The most popular technologies - group 0-50

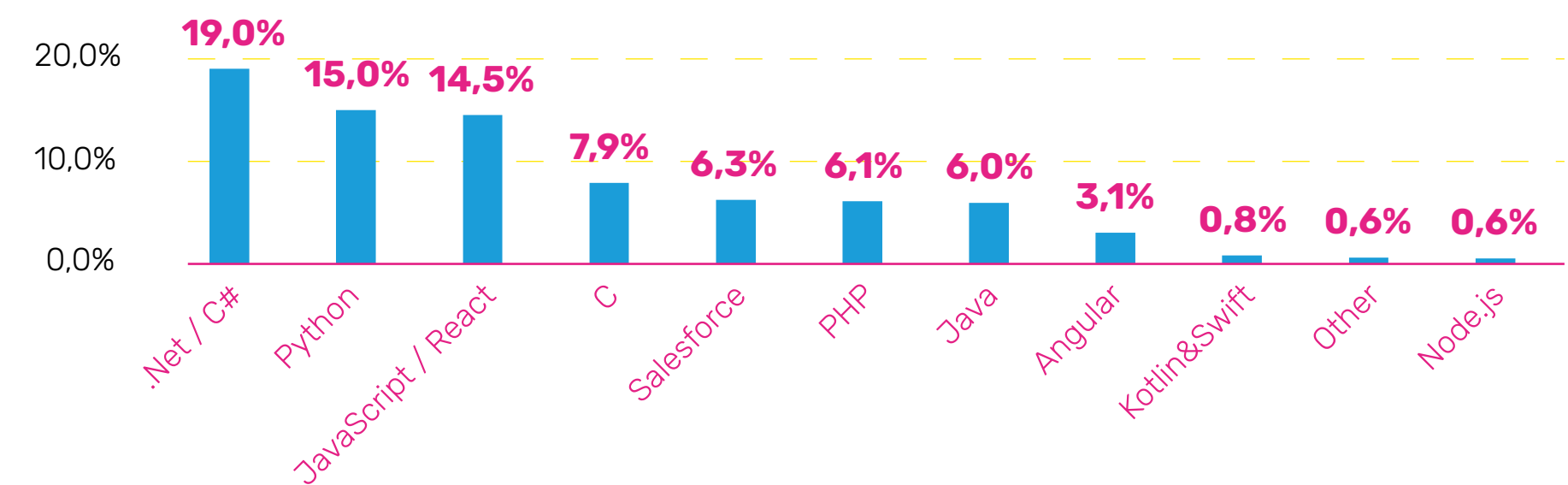


In the case of micro and small companies, JavaScript is a clear leader. Respondents' answers indicate that approximately 20.4% of all IT specialists within the surveyed companies are familiar with this technology.

Java came second with 18.5% of the total number of specialists. Such a result is not surprising, given the current market demand for programmers specialized in these technologies. JavaScript is currently the most popular technology not only in Poland but also in the world.

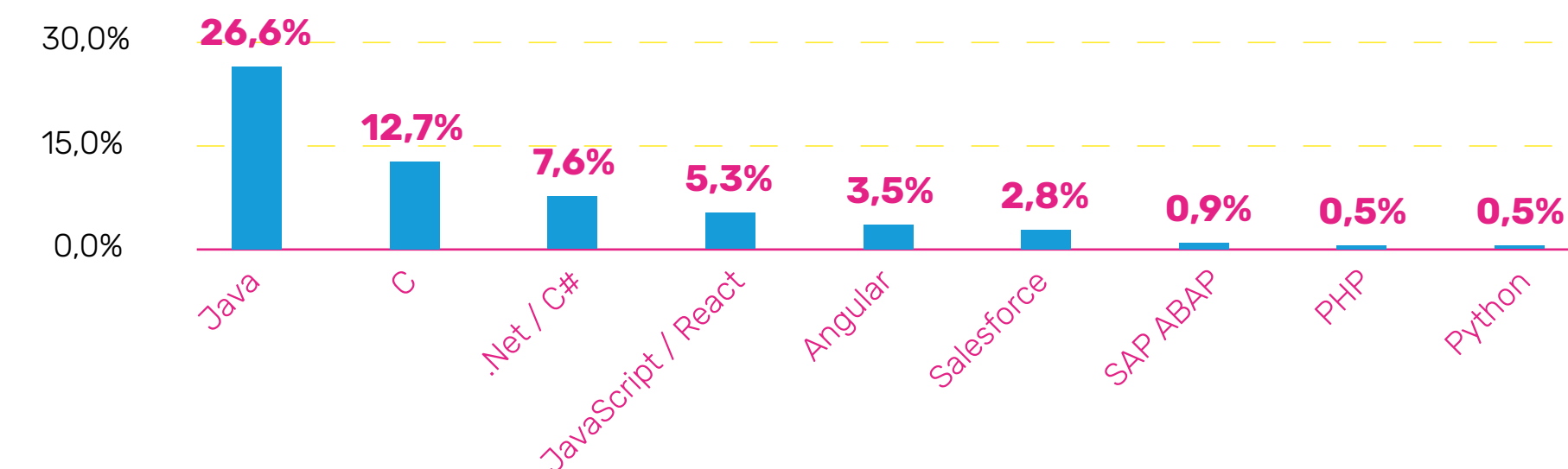
.Net is the most popular in a given group (with 19% awareness). However, for companies with a size of 50-250, the high popularity of Python (15%) is noticeable, as the demand for these specialists is currently high in Lodz for companies from the 50-250 group and, at the same time, extremely low for other companies. JavaScript / React comes in third with 14.5% knowledge.

The most popular technologies - group 50-250



In companies with more than 250 employees, Java is the clear leader in technologies, with declared knowledge of just over a quarter of specialists working in the surveyed companies. Then, the knowledge of languages in the combined group for C languages is demonstrated by 12.7% of specialists, and 7.6% belongs to .Net / C #.

The most popular technologies - group 250+



* Source: City of Łódź + Grafton Survey. We determined technologies' popularity based on the participation of specialists declared by the respondents as people familiar with a given technology and the total number of IT specialists in companies answering the question.

Even though the past three years have been difficult for many companies worldwide, large numbers of businesses were able to adapt to new circumstances, and – as an outcome – the current job market is still very demanding for recruiters and Hiring Managers. The Polish market is constantly struggling with specialist shortages, especially in the IT domain in many areas, such as software development.

Even though JavaScript is currently one of the most popular technologies used globally within the software development area, it is interesting to observe different trends for specific regions and companies' profiles, which gives us a better understanding of our job market. For example, as the research conducted by Grafton Recruitment has shown (the above mentioned), JavaScript is the leading technology only for the micro&small companies (1-50 employees) in Lodz while for medium companies with 51-250 employees – Python and large companies – 250+ employees – Java. The difference can be easily explained after having a closer look at our market, and the business concluded here as the technology must be fitted to the products developed.

Smaller companies in Lodz focus mainly on web-based products, for which JavaScript was the first choice due to its specifics and capabilities. At the same time, Python, for instance, could not be as practical (which is why its usage level is relatively low for this group of companies). Therefore, different kind of products, as well as their complexity created by the medium and large companies that operate in Lodz, result in other technology selections. That is why we can observe the more extensive popularity of Python and Java in this market.

Looking at the previous years and companies' growth plans in Lodz, we can assume that the leading technologies for this sector will remain on the same level, and no significant changes on the top list should be expected in the incoming months.

Agata Jemioła

*Branch Manager
Grafton Recruitment*



IT specialists in the employment structure

The study also showed differences between the structures of technical departments in small, medium and large companies. Companies employing up to 50 employees, as well as those from the 50-250 group, have a similar percentage share of technical specialists in the entire employment structure (70.5% and 69.5% respectively). In companies employing over 250 employees, the number of technical specialists is the lowest, i.e. about 58.1%. The number however is significantly higher when considering within the group IT Companies only - 68,1% of the workforce are IT Specialists in that case.

	Total number of employees (Survey Data	Number of IT specialists (Survey Daya)	% of IT Specialists (TOTAL)	% of IT Specialists (IT companies only)
0-50	569	401	70,5%	69,8%
50-250	2128	1479	69,5%	72,5%
250 +	11 997	6 972	58,1%	69,1%
Total	14 694	8852	60,2%	68,9%

Using the data from the report, there has been an estimate prepared using known additional 62 entities from the region grouped into similar categories as survey companies. The analysis has concluded that for chosen entities the estimated number of technical workers is around 13 615.

	Number of Technical workers (Survey Data)	Estimate of number of technical workers	Survey data + Estimate
0-50	401	575	976
50-250	1479	1629	3108
250 +	6972	2559	9531
Total	8852	4763	13 615

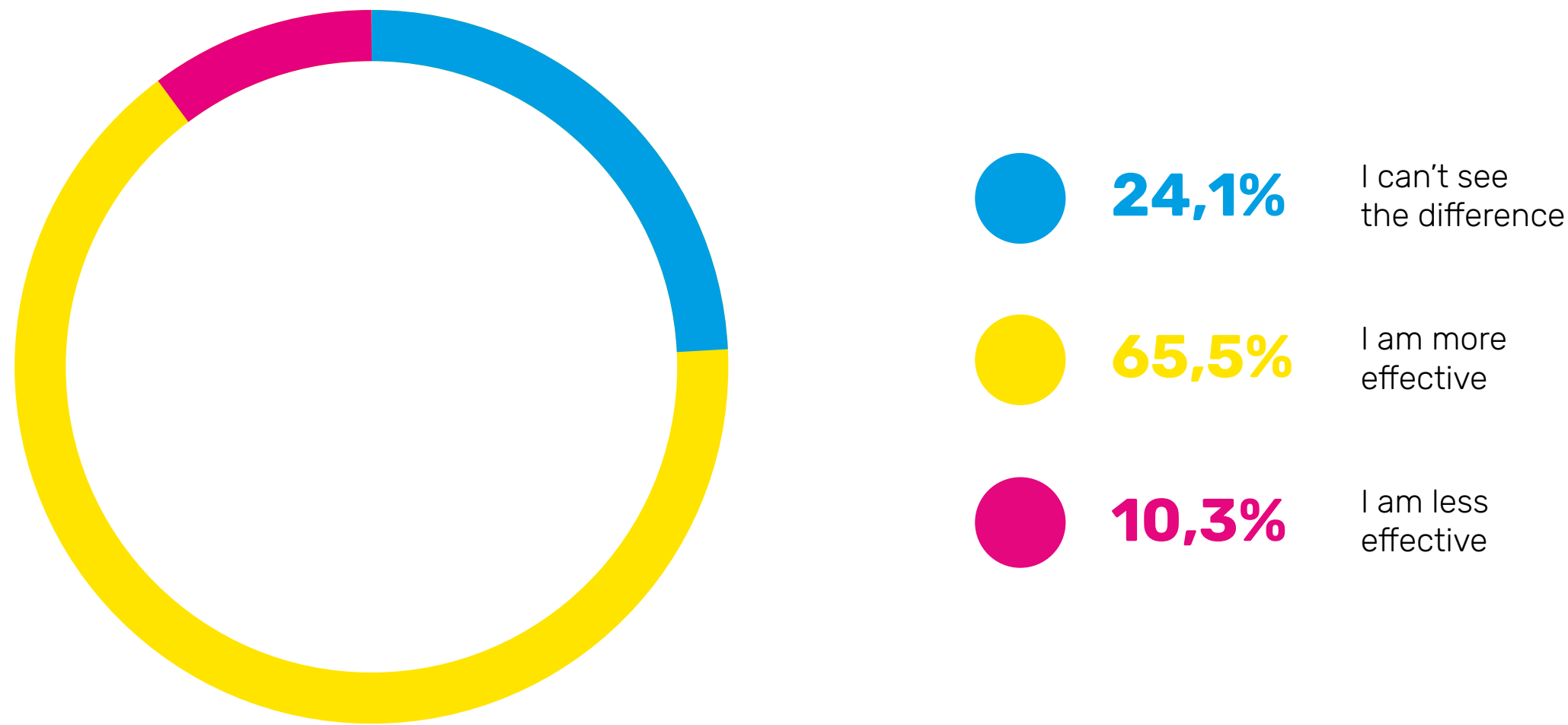
*** the estimate assumes 62 additional entities for which the number of technical employees is the median number of technical employees resulting from the responses from the survey for companies of appropriate sizes (0-50, 50-250, 250+). Most representative companies from each category were chosen for more accurate results.*

Remote work in the opinion of IT specialists

In recent months, we have observed a strong trend of returning to offices from remote work, also in the IT sector. To find out the opinion of IT employees on this formula of work, we asked them to subjectively evaluate the effectiveness of working from home in the survey.

Sixty-four respondents from all over Poland answered this question. The survey shows that 65.5% of IT specialists consider their work at home more effective than in the office. But, interestingly, every 4th person doesn't notice the difference.v

The effectiveness of remote work as assessed by IT specialists



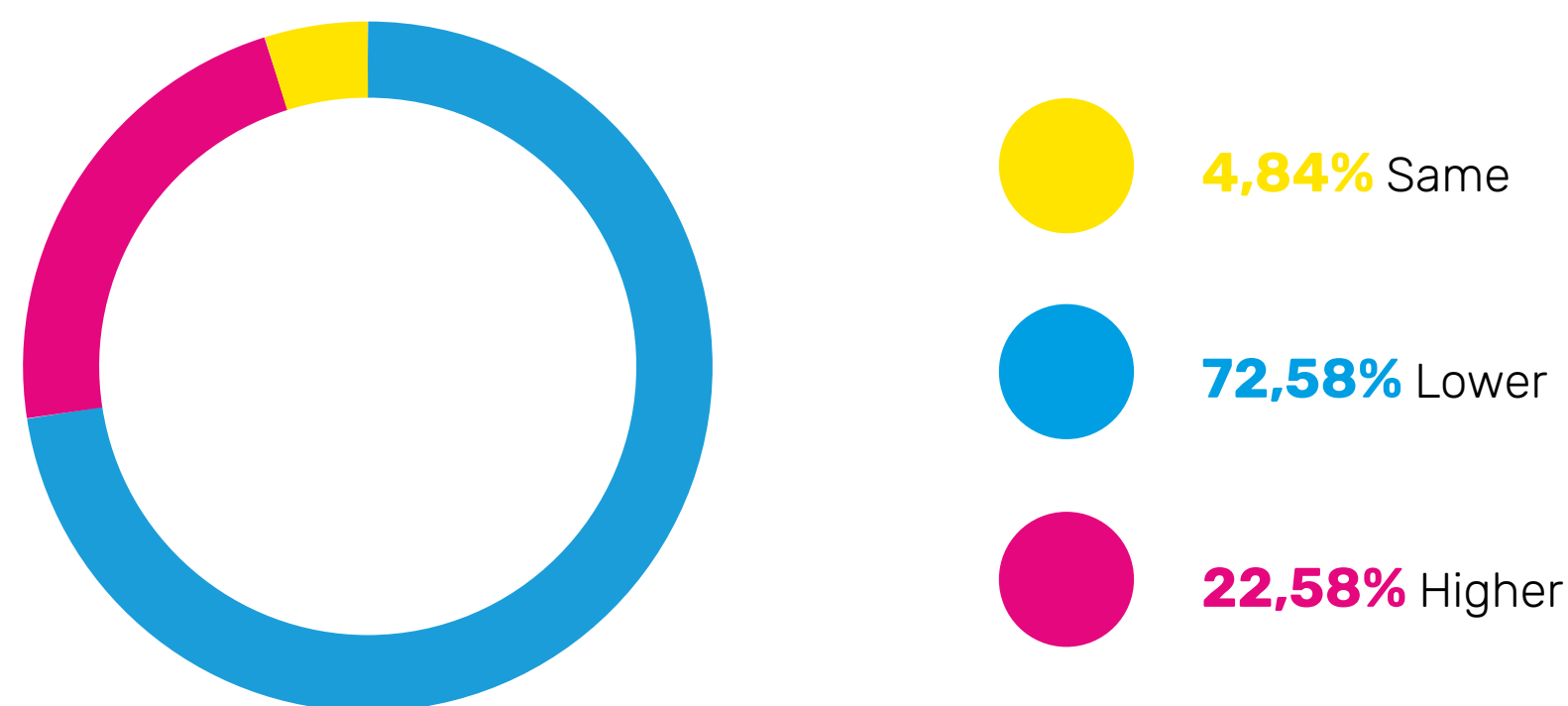
A survey conducted by Grafton Recruitment n = 338, survey date 02/2022, IT specialists, LinkedIn.

Remote work in the opinion of IT specialists

We also asked companies in Lodz that employ IT specialists about the subjective assessment of the effectiveness of remote work. Comparing their responses with those of employees showed that employers are much more likely to be inclined to the position that remote work is less effective. Only less than 5% of employers noticed higher effectiveness of remote employees. However, in employee surveys, the answer „I am more effective working remotely than working from the office” was the most frequent.

IT specialists from all over Poland took part in the study. At the same time, the employers’ perspective was represented only by companies based in Lodz. Therefore, we need to consider a certain level of discrepancy when analyzing these data. Positive assessment of the effectiveness of remote work by employees may result from a subjective evaluation of the implementation of tasks. For example, many employees associate the transition from in-office to remote work with eliminating distractors characteristic of the office environment. This association could have given the apparent feeling of having more time during the working day and the illusion of completing tasks faster. In the case of employers, answers could be given not only based on unstructured observation but also internally conducted objective measures of employee effectiveness.

The effectiveness of remote work as assessed by Lodz IT companies



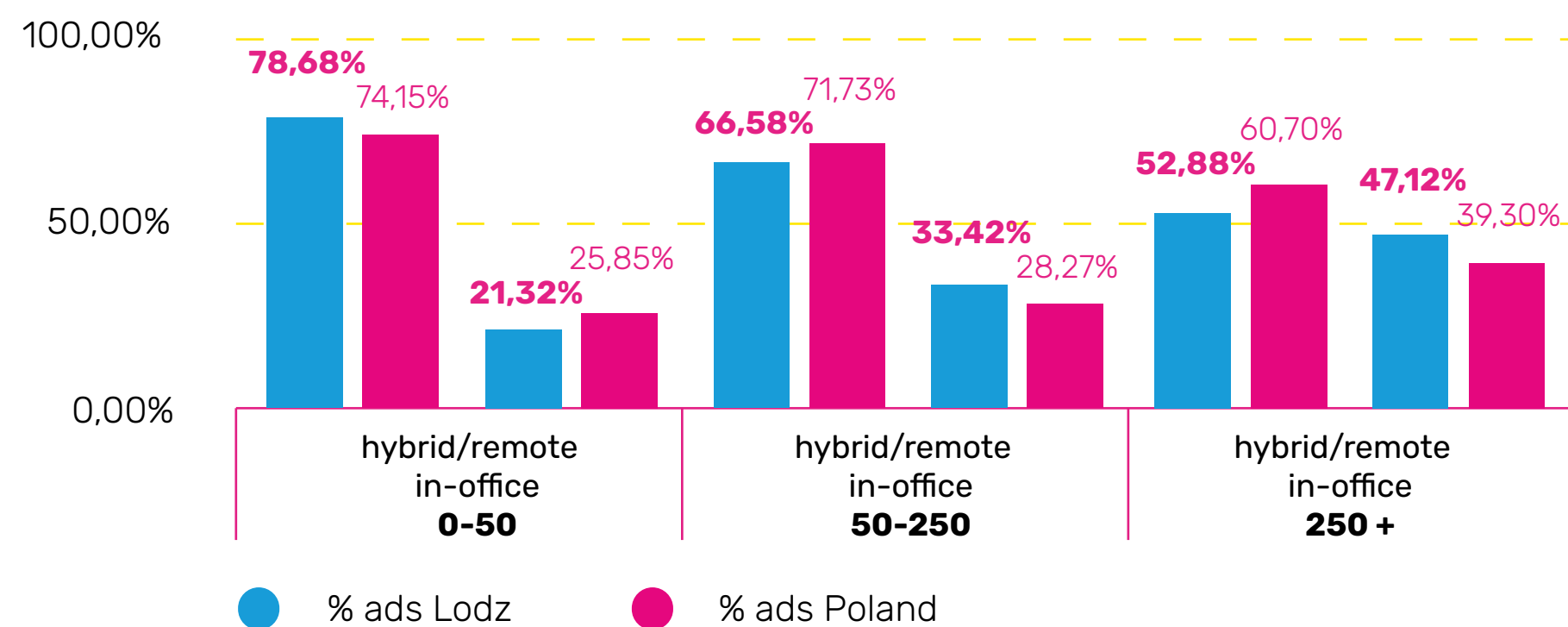
A survey conducted by Grafton Recruitment n = 62, survey date 12/21 - 03/2022

Hybrid/remote and in-office work – Lodz vs Poland

The analysis of job offers published on specialized IT portals shows that Lodz IT companies are more likely to provide in-office employment than entrepreneurs from other cities.

The differences that prevail in favor of remote or hybrid work in the case of Lodz are noticeable, especially in the case of companies with employment from 0 to 50 employees. Lodz employers are more likely to offer remote jobs. The number of advertisements with the option of working in a remote or hybrid formula is 4.5% higher than the benchmark (all job adds in Poland).

IT job adds - Lodz vs Poland (hybrid/remote work and in-office work)



On the other hand, in the case of companies in the 50-250 and 250+ size ranges, we notice a slightly lower willingness to offer remote or hybrid work opportunities. In the case of the 50-250 group, we see over 5% more job advertisements with a formula that allows at least some remote work. It is even more visible in the 250+ group, where we observe a 10% difference in favor of the in-office position.

Grafton data source. Analysis of popular portals with IT job offers.
The analyzed period is September 2021 - August 2022.

Hybrid, remote or in-office? Lodz IT market after pandemic

The development of the Lodz IT market in terms of the forms of employment offered is in line with the nationwide trend of returning from remote to stationary or hybrid work and even distinguishes itself from the rest of the country by a greater tendency to offer employment in the latter formula.

Our research confirms that medium-sized and large companies often offer stationary or hybrid work in Lodz, and smaller companies rely on the remote formula. There can be many reasons for this distribution of results. 250+ companies are most often highly recognizable in Lodz. Most of them have already built a strong brand as employers, which may be a factor in increasing the attractiveness of employment. On the other hand, a significant number of companies with employment at 0-50 do not exist so strongly in the minds of the inhabitants of Lodz. When competing with big players, they must offer the right conditions to attract new talent. One of them is the possibility of entirely remote or hybrid work.

Large organizations also tend to be more diverse in talent; horizontal and vertical shifts are common. By treating internal promotions as one of the ways to fill vacancies, they don't have to fight for specialists as hard as small IT companies.

In turn, it is easier for small organizations to offer their employees hybrid/remote work because managing a smaller team generates fewer communication disruptions than in larger structures. Another factor to consider in our research results is the business profile of IT companies in particular groups. For example, the most significant number of enterprises dealing with creating and developing embedded systems software in Lodz has over 250 employees. Working on this solution involves physical contact with the equipment, often obliging employees to visit the company's premises. To meet this challenge, some organizations allow employees to take equipment home to enjoy the benefits of remote work without sacrificing efficiency. However, this is a relatively rare solution in the IT market, so the advantage of stationary job offers in large IT companies in Lodz may result from an objective necessity.

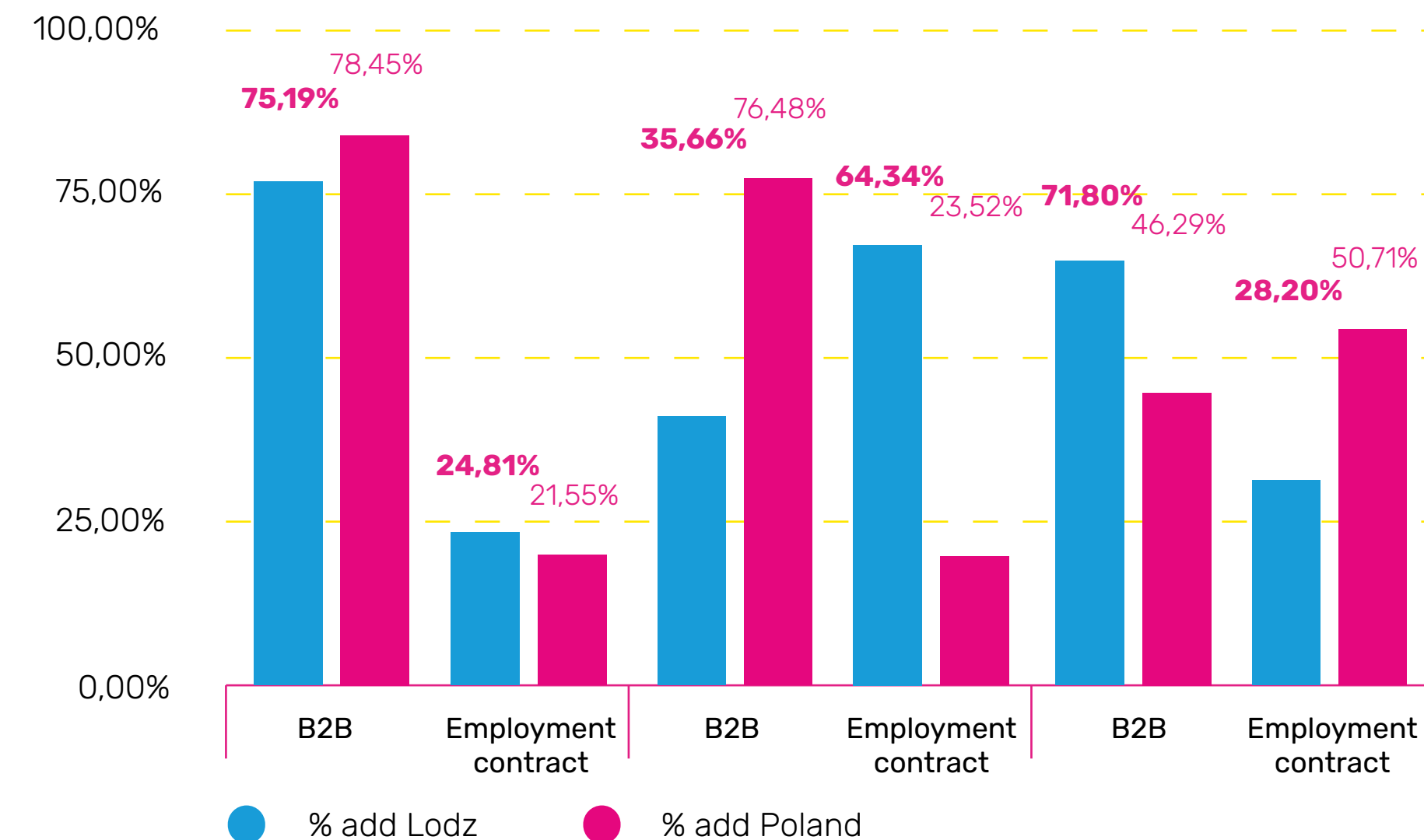


Forms of employment

– Lodz in comparison to Poland

The Lodz IT job market analysis shows one more trend: the growing popularity of freelance work and B2B cooperation. The chart presents a percentage comparison of the number of jobs adds in IT, offering employment under an **employment contract**, and the possibility of employment in the B2B formula. We have divided them into three categories.

Employment forms offered in the IT sector in Łódź and all over Poland (B2B vs. Employment contract)*



Companies with 0-50 employees and those from the 50-250 group decide on B2B cooperation less often. Although at the same time, the trend in the 0-50 group is very close to the national average, the difference between the proportion of advertisements with the possibility of contract cooperation is only 3.3%. On the other hand, we notice significant differences in the 50-250 group. Companies of this size often offer work only in a formula based on an employment contract than the average in Poland. We observe that only 35.66% offer B2B options compared to the national 76.48%, which means that companies in this size of employment prefer contracts that are generally considered more stable from the employer's perspective.

Contrastingly, in the 250+ group, we see the opposite. This time, companies from Lodz are much more flexible in cooperation with contractors compared to the Polish average. It is an exciting observation for reasons related to the apparent general preference of large companies to offer the formula of cooperation in a form that is the most constant and stable. Furthermore, the willingness to cooperate in the B2B formula of the most significant players in the Lodz IT market shows the labor market's dynamics and competitiveness in attracting the best talents.

* The B2B category also includes optional B2B employment (some job advertisements offer a choice of form of employment).

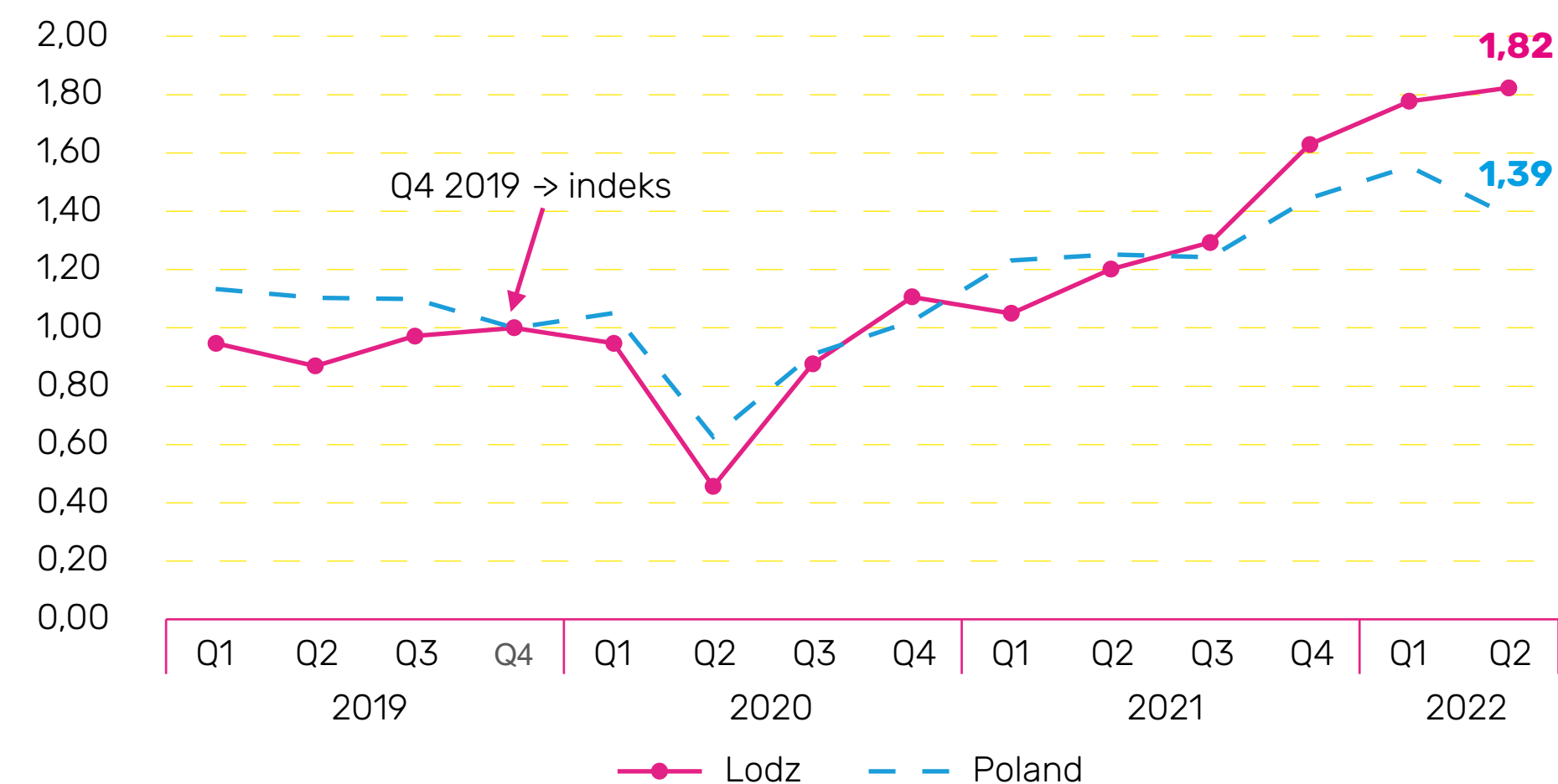
Demand for IT candidates

We measured the demand for candidates in the IT market by analyzing the number of jobs offers posted on a selected recruitment portal from 2019 to the end of Q2 2022 and by indexing the value in Q4 2019.

As for many industries, the pandemic strongly affected the IT sector. However, the situation picked up after a temporary drop in demand for specialists in the second quarter of 2020. From the third quarter of 2020, we observed a steady increase in IT staff demand.

The revival of the IT job market in the Lodzkie Voivodship is greater than the Polish average in almost every measured quarter. In recent months, the demand for IT specialists has grown dynamically, and finally, in the second quarter of 2022, it overtakes the benchmark for the whole of Poland by 43 percentage points.

Demand for IT specialists - Poland vs the Lodz Voivodeship (Q1 2019 - Q2 2022) - index = 1 Q4 2019



Demand vs supply of technology

Among the companies operating in Lodz, a large variety of projects is observed, and thus also the tools used to implement them. In this group, however, we can distinguish several most popular technologies, both in demand and supply of specialists. These include .NET, C ++, cloud technologies, SAP, and Java.

In our research, Java is one of the technologies most often used by small, medium, and large enterprises. Therefore, it proves the massive demand for candidates specializing in this language.

The situation is slightly different in the case of SAP. Here, we observe a small supply of specialists with a simultaneous increase in demand, which may mean recruitment difficulties and the need to use the resources of other markets. The same applies to cloud technologies: although they are gaining in popularity every year, we still have a shortage of specialists in this field. The growing demand for this technology may also translate into great recruitment difficulties. As the ever-increasing demand for specialists in this area is already visible throughout the country, it poses a challenge for Lodz entrepreneurs.

In addition to the technologies above, the demand for Python and JavaScript specialists, used mainly by employees of companies with 50 to 250 employees, has been growing for several years (JavaScript was also the most frequently mentioned language among companies employing up to 50 employees). At the same time, in the first quarter of 2022, in Lodz and other Polish cities, we observed a slight slowdown in the current increase in the number of job offers in this area. With this tendency, the risk of an outflow of specialists from Lodz to other regions of Poland is decreasing. This situation can be a good recruitment opportunity for newly-opened businesses.

CBRE

CBRE Workplace Advisory Interview Report



Undoubtedly, the pandemic accelerated the organization’s efforts to implement changes in the organizational culture and new ways of working to remain attractive in the market. These changes also apply to the IT sector. The nature of work that allows you to perform tasks from anywhere opens the possibility of obtaining work supra-regional. The expectation of greater flexibility and care for the employee’s well-being constantly changes the industry’s landscape and relations within the organization. We also see what is observable in the Polish market locally in Lodz.

We present the report on qualitative research, in which we create a set of critical changes covering the IT market in Lodz. Thanks to the organizations’ leaders who participated in the survey, we see what factors are transforming regarding employee expectations towards development opportunities and flexibility. We also look at how organizations respond to these changes through appropriate policies and actions in cultural and spatial areas.

Agata Mróz
*Associate Workplace Director, Workplace
Strategy & Change Management,
CBRE*



Work place strategy CBRE research [methodology]

This qualitative study was conducted by the CBRE Workplace Strategy & Change Management Team in Q2 2022. As part of the study, seven detailed expert interviews were conducted with leaders of IT industry organizations operating in Lodz. The surveyed companies include, among others: software houses, logistics centers, IT outsourcing, and branches of international organizations of origin: e.g., American, Ukrainian, and Polish enterprises. In addition, the sample includes organizations that:

- functioned before the pandemic
- started operating during (or at the end) of the pandemic.

The interviews were supplemented with a short questionnaire to examine the phenomena defined by the CBRE Workplace Strategy & Change Management Team as crucial for the work environment in the post-pandemic reality. The survey results were presented in the relevant thematic modules. For methodological reasons, the questionnaire was completed by six people.

The IT market and the City of Lodz Office

The study participants described the attitude of the City of Lodz Office in maintaining relations with business and the role in attracting business to the city as active and influential. It was underlined that:

- More than offices in other large urban centers, the City of Lodz cultivates relations with companies and is closer to business
- The City of Lodz Office conducts effective marketing and effectively provides information and assistance to investors looking for a place to set up a company seat.
- Companies have access to the structures of the City Office, unlike in the cities such as Warsaw or Wrocław. The market there is more extensive than in Lodz, and thanks to it, the City of Lodz Office can take good care of organizations running their activities here.

The City of Lodz Office looks after companies and is much closer to business than the offices in other large urban centers.

Recruitment strategies

Recruitment

It is estimated that some candidates in Lodz are more passive in creating and expanding their professional social network than candidates in other parts of the country. This challenge is responded to through promotional activities such as the organization of webinars, partnerships with universities, internships, and marketing campaigns in the city.

Candidates from Ukraine and Belarus are assessed as highly qualified and desired specialists on the market. In addition, before and during the pandemic, all interview participating companies declared openness to employees from abroad, including Ukraine and Belarus.

Candidate profiles

Companies differ in their approach to the requirements regarding the profiles of candidates/portfolios.

In Software Houses, in addition to indicating specific specializations (developers with domain skills, testers, business analysts, programmers, coders, solution architects, IT Engineers, UX designers, and people with experience in working in industries supported by the company), they are also looking for people with soft skills, such as camaraderie, highly developed communication skills or a sense of fit with the company.

Changing recruitment strategies affected by a pandemic

The pandemic period affected not only the level of retention in the organization but related challenges with retaining talent. For many companies, including companies participating in the survey, this period also became the moment of decisions to change the current recruitment policies.

The pandemic became the moment for decisions to change the current recruitment policies. It opened the possibility of acquiring people from outside the direct location of the company's headquarters

The pandemic increased the popularity of remote work. The popularization of the home office form (often considered equivalent to work performed from an office) has opened the possibility of recruiting employees outside the company's direct location. As a result, working in the office ceased to be one of the main determinants of formal membership in an organization or cooperation with it. These trends are also visible among the companies participating in the survey. We observe the extension of recruitment strategies across Poland and the acquisition of people from outside Lodz without the relocation requirement. The trend is also visible in organizations previously giving preference to employment contracts.

Post-pandemic work model

Work model - hybrid work

Due to the growing popularity of remote work, companies that previously had an office plan to introduce a hybrid work model or have already introduced such a model. Organizations taking part in the study focus primarily on the flexible model. A model where decisions regarding access to the home office are made within the teams.

During individual interviews, those models transpired.

Model on request, where decisions will be made in departments, taking into account at the same time that not everyone can or has the conditions to work remotely, and it depends on the assessment of predisposition and the employee's living conditions. There is an internal discussion about whether to introduce the possibility of working from anywhere.

Flexible model. Striving for office occupancy at the level of 60% of the total headcount every day. Using incentives, but not obliging you to come to the office. They are making decisions at the level of smaller structures.

Full Freedom. The possibility of using the office, the option of remote work.

Remote work has become one of the key attractors for the IT industry. Although different in the analyzed organizations, the development of hybrid work will be based on choice. With the possibility of working from home or/and from the office. Flexibility without being forced to come to the office.

Hybrid model. Look at the office from a different perspective. Coworking and a network of offices where you can quickly reach, work and continue working as a home office. A place to work is a workstation, a home, and the office will be a meeting and event place. This model offers people the opportunity to experience in the office what they cannot experience at home. It strives for people to appear in the office with an ideal target model: 50% office, 50% remote work, and no coercion.

Companies don't want to force their programmers to come to the office; instead, they focus on using incentives, e.g., integration actions.

Key attractors [types of contracts and remuneration] according to WPS CBRE Expert Interview Report

Remuneration

The costs of employing IT specialists in Lodz are slightly lower than in other cities in Poland. At the same time, it is emphasized that an attractive salary is a crucial factor for candidates. In the post-Covid era, where specialists are looking for employment without geographic restrictions, the difference in remuneration is not and cannot be grossly lower compared to other cities in Poland to attract talent. The type of contracts concluded is closely related to the company's business strategy, development goals, office strategy, and the programmer's role in the organization.

Type of contractors and business strategy

Companies with an advantage of *Employment contract* type of cooperation: In companies where the majority (80% or more contracts) employ an *Employment contract* employment policy, long relationships with employees prevail, care for maintaining low turnover, the importance of the sense of belonging to the organization, and stability of employment are emphasized. In the interviews, it was underlined that the organizations employ people with families, i.e., looking for stabilization and expecting companies to be highly active in the benefits offered, also for the families of employees. Therefore, in the post-pandemic realities, it is planned to keep *Employment contract* as the dominant form of employment.

Companies with hybrid contract mode: In organizations with a hybrid model, it is emphasized that employees have a choice of the type of contract. The most common candidates are young people (around 25). An element of the candidate acquisition strategy are activities focused on activating students of critical faculties for the industry. However, these organizations indicate that in the post-covid reality, they experience challenges related to retention and reduced candidates' loyalty.

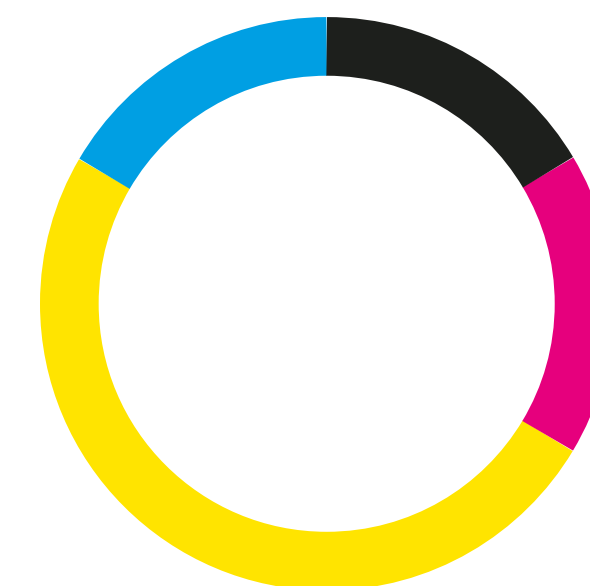
Impact of the pandemic on employer-employee relations

Companies that offer forms of employment primarily or predominantly based on a *B2B or B2B + Contract* for a specified task type of cooperation indicate that the post-pandemic period is a time of increased

rotation in the entire industry, decreased loyalty, and a sense of ties with the company. The most significant risk of leaving is associated with the best specialists and people who are briefly in the company. In organizations that offer almost exclusively *Employment contract* type of cooperation, it is generally pointed out that rotation is not such a strongly felt phenomenon. According to some, this type of employment increases employees' sense of belonging to the company.

A decline in employee loyalty was signaled primarily among organizations with an employment model based on B2B and *Contract for a specified task*. In companies where *Employment contract* prevails, this type of contract increases the feeling of belonging to the company.

How has the meaning of employment based on a *Employment contract* changed (in your opinion), comparing the situation before and after the pandemic?



- Much less important than before the pandemic
- Less important than before the pandemic
- I don't see a significant difference
- More important than before the pandemic

The survey results indicate that the type of cooperation based on *Employment contracts* under the influence of the pandemic is currently in a stable position. For the most part, it has retained its importance and has slightly gained in importance.

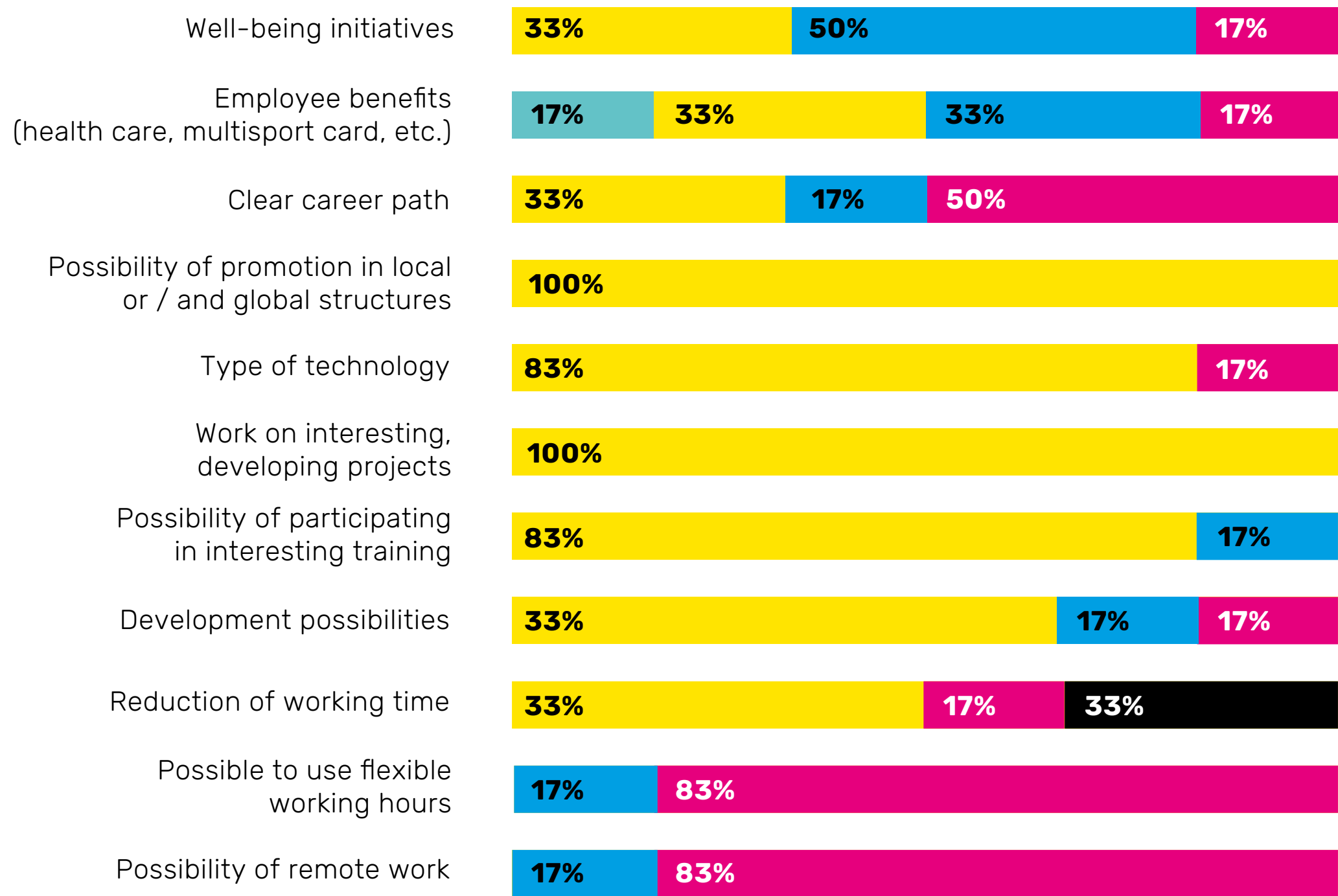
Key attractors [offer for candidates and benefits] according to WPS CBRE Expert Interview Report

It is estimated that the benefits offered to employees of IT companies in Lodz are pretty even. The key differences between the benefit packages offered are specialization, type of projects, office location, and approach to flexibility. As a result, intense competition is expected between companies in employee benefits quantity and quality.

The key benefits/elements that distinguish the offer include:

- **types of technology and exciting projects**
- **clear development path and training offer.** It is emphasized that an essential factor influencing employee rotation is the need to develop competencies, which is reflected in the search for new technologies and new projects. It affects the readiness of developers to build their own path by those who don't receive a clear development perspective from the company. Development programs for specialization, specialist training, and support in the event of a change of career path are indicated as ways to care for development.
- **flexibility and availability of remote work.** At the same time, it is indicated that companies are making additional efforts to encourage employees to come to the office due to the need to integrate teams and create and maintain a creative atmosphere. Some solutions offer total freedom in terms of the place and hours of work. Such flexibility translates into additional paid leave, access to co-working zones or offices at any time, an attractive location, and access to municipal infrastructure supporting social life.
- **wide range of social benefits** (catering, integration, family, sports, medical care, etc.)
- **attractive office space.** It is underlined that high standard ergonomic space standard, with good access to public transport, with co-working, integration, and social infrastructure, as well as supported in these functions by the city infrastructure.

How has the meaning of the following items changed for your employees comparing the pre-pandemic and post-pandemic time?



- Much less important than before the pandemic
- Less important than before the pandemic
- I don't see a significant difference
- More important than before the pandemic
- Much more important than before the pandemic
- I don't know / lack of sufficient information / not applicable

Pilot study supplementing the qualitative part in the form of a questionnaire, number of respondents: 6



Lodz office space market overview



Lodz – on the wave of the development of the IT and Business Services sectors

Lodz’s strengths and business potential allow us to forecast that, shortly, the city will achieve a leading position in the country in terms of investment attractiveness. In the last few years, we have observed many changes that have allowed Lodz to transform into a modern metropolis, offering attractive living and business conditions.

Flexible space is gaining more and more importance for the real estate market in Łódź, and the number of co-working spaces is systematically growing, which is a beneficial solution for new foreign investors. Furthermore, given the high inflow of investments to Lodz, especially IT companies from the East (Belarus and Ukraine), flexible solutions significantly facilitate the planning and development of operational activities.

We are observing a dynamic increase in the business community’s interest in Lodz, especially concerning foreign investors. Lodz is quickly gaining popularity. In recent years, this has been the case in Katowice and Gdansk, where many inquiries and projects regarding expanding business processes have been received. There are more and more arguments that will put Lodz on the next wave of new investment and business projects. Apart from the development of infrastructure, the key advantage of Lodz is that the city is becoming a fashionable place to live and work.

The post-industrial character of the city ceases to be a burden and becomes a distinguishing feature and attractor for both labor market participants and business organizations – employers. Similarly, the proximity of Warsaw is no longer a challenge and becomes an opportunity for Lodz. In the era of hybrid work, Lodz can benefit from the vicinity of the capital while building its position with lower costs of running a business and living costs, with the city’s attractiveness comparable with its multi-functional social and business offer.

The growing interest of foreign investors in Lodz has been visible for the last two years, while the peak of popularity as a leading location for the BSS sector is yet to come.

Jan Banasikowski
Associate Director,
Advisory & Transactions,
CBRE



The future of office and building space according to WPS CBRE Expert Interview Report

Building features

The office and the building can be one of the distinguishing features of the local office market due to the convenient location, stunning view of the panorama of Lodz, design, or, for example, access to services such as a gym and fitness club. The role of an attractive office location (next to Manufaktura) was emphasized in integrating the team and organizing initiatives to encourage employees to come to the office.

The plan of the City of Lodz supports the idea of easy access to co-working zones. This concept is implemented through a network of co-working offices available from various locations, which respond to the needs of employees regarding the need to work 2-3 hours in a shared, collaborative space. In addition, sometimes employees get access to a remote home-office internet platform, home office, which provides access to co-working space.

Office functions

In the hybrid model, the office changes its function. It is to be a space optimized for the organization's needs, a meeting place, community building, information exchange, and creativity, as well as a workspace for those who do not have conditions for remote work at home.

Companies based on *Employment contract*

An essential aspect for the interlocutors is the ability to change the office function to one that supports the hybrid work mode, team integration, and creativity and builds communities. In the future, this can be achieved, for example, by arranging the office - attractive spaces for creative meetings, screens, interactive boards, rooms for remote sessions, places for online meetings, and rooms for meetings with online managers. Additionally, there was a departure from an open space with several desks in favor of an area with rooms for 2-3 people and a well-lit and attractively arranged office.

Other modifications include

- changing the office function and shifting proportions by increasing the space available for collaboration
- introducing the concept of 15 minutes to the office, which is favored by the infrastructure of Lodz
- providing employees with co-working offices in various parts of the city
- developing the hot desk/desk sharing system with temporary workplaces.

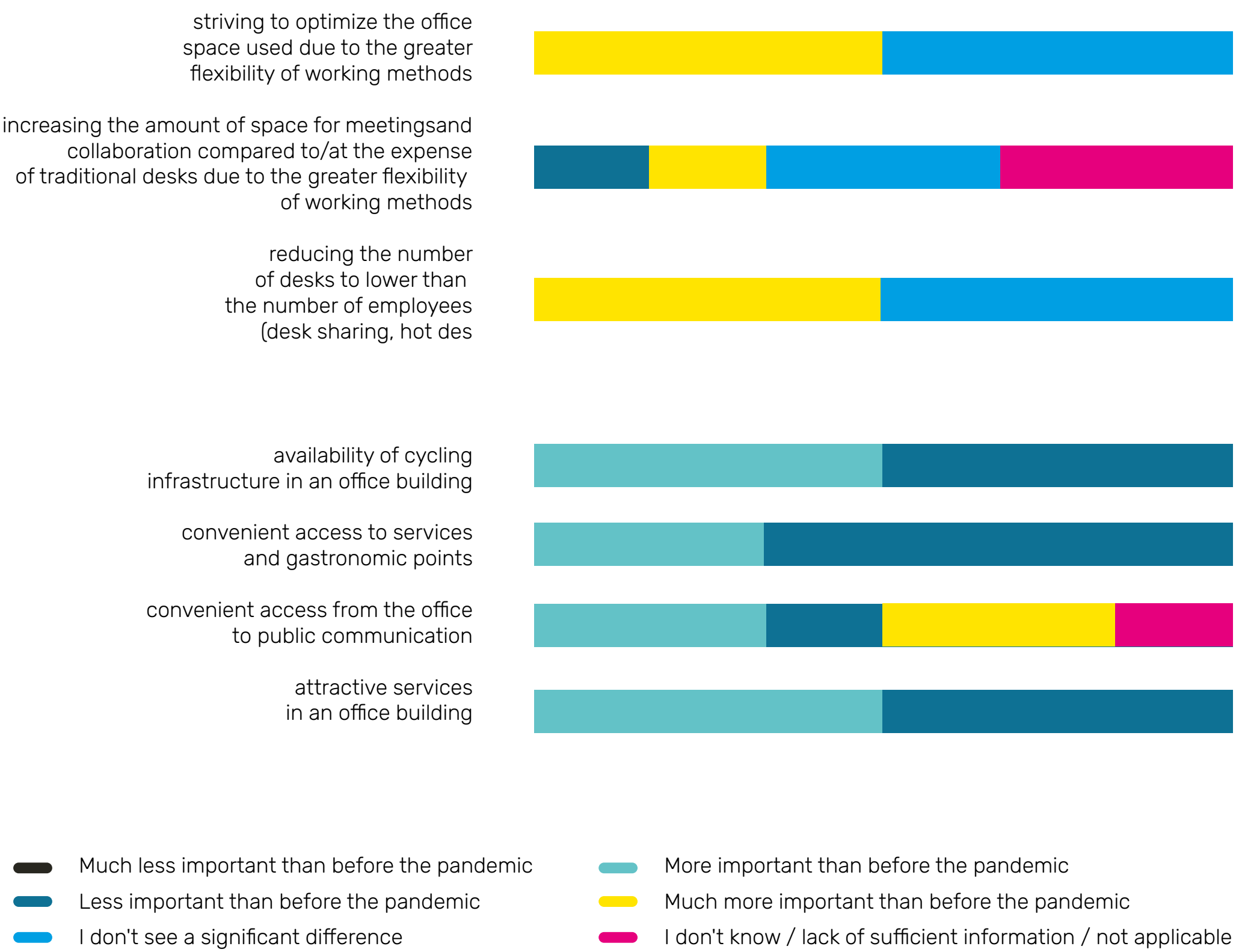
Companies operating in a mixed model

Representatives of companies employing its specialists within the mixed (*Employment contract* + B2B) model with a predominance of *Employment contract* indicate that they want to actively monitor and create a market standard and offer the opportunity to work in the office to people who are interested in this mode of work. Hence, it was decided to rent co-working offices in several cities (including Lodz) for people who previously worked 100% remotely.

Companies operating in the B2B model

Where the business model is based on B2B contracts (and the company does not currently have an office), it is planned to use the co-working offer. Such a space supports the organization of brainstorming, meeting places, and team building and provides office space for employees.

How has the meaning of the following items changed for your employees comparing the pre-pandemic and post-pandemic time?



The survey results confirm that the office of the future will change its function due to the widening of the range of possible places to perform tasks. The building infrastructure did not gain, but it did not lose its importance due to the pandemic.

An interesting thread remains the issue of reasonable access to the office via public communication - opinions are divided, but according to the leaders participating in the survey - with a slight indication of the increased importance of this aspect in specific cases.

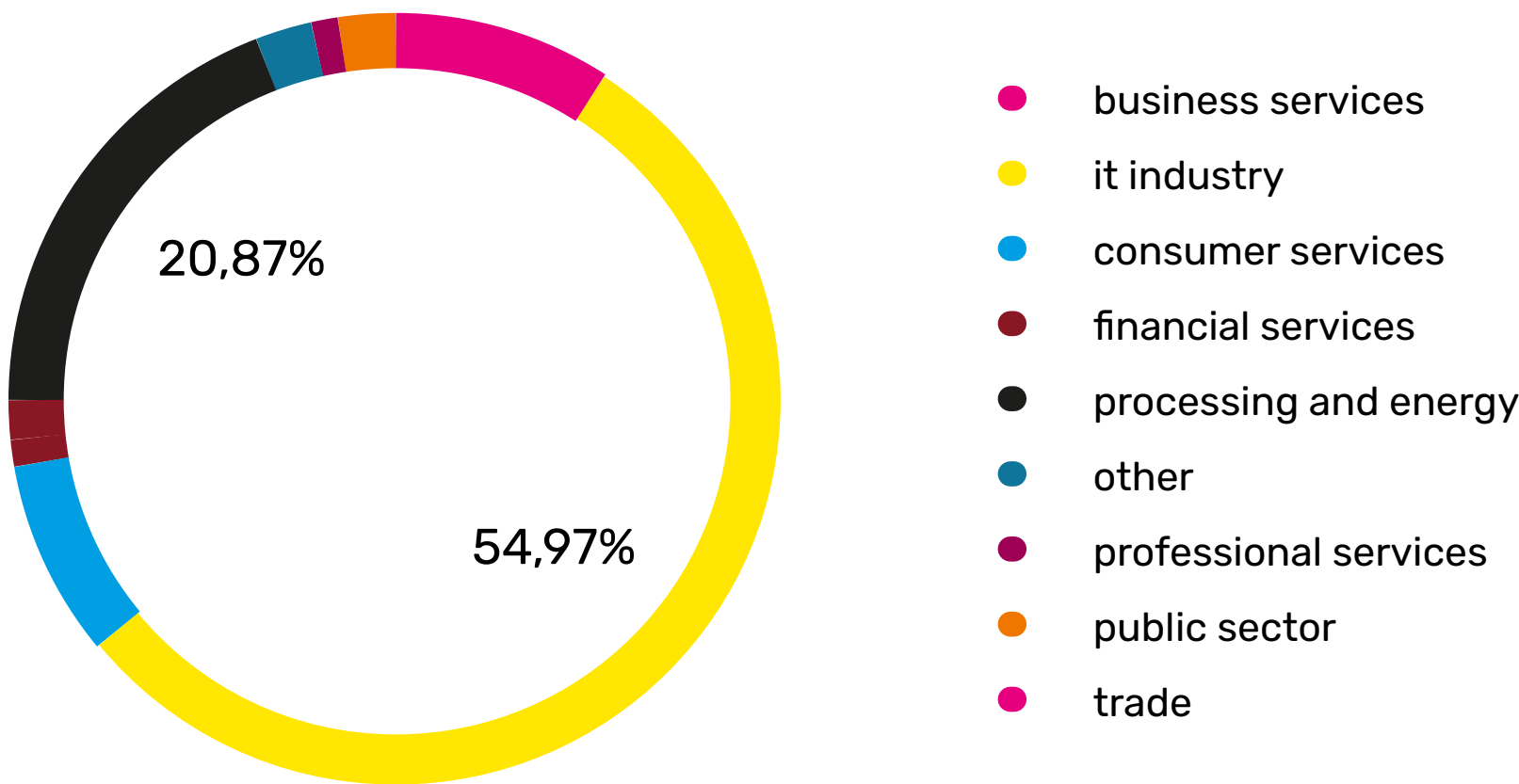
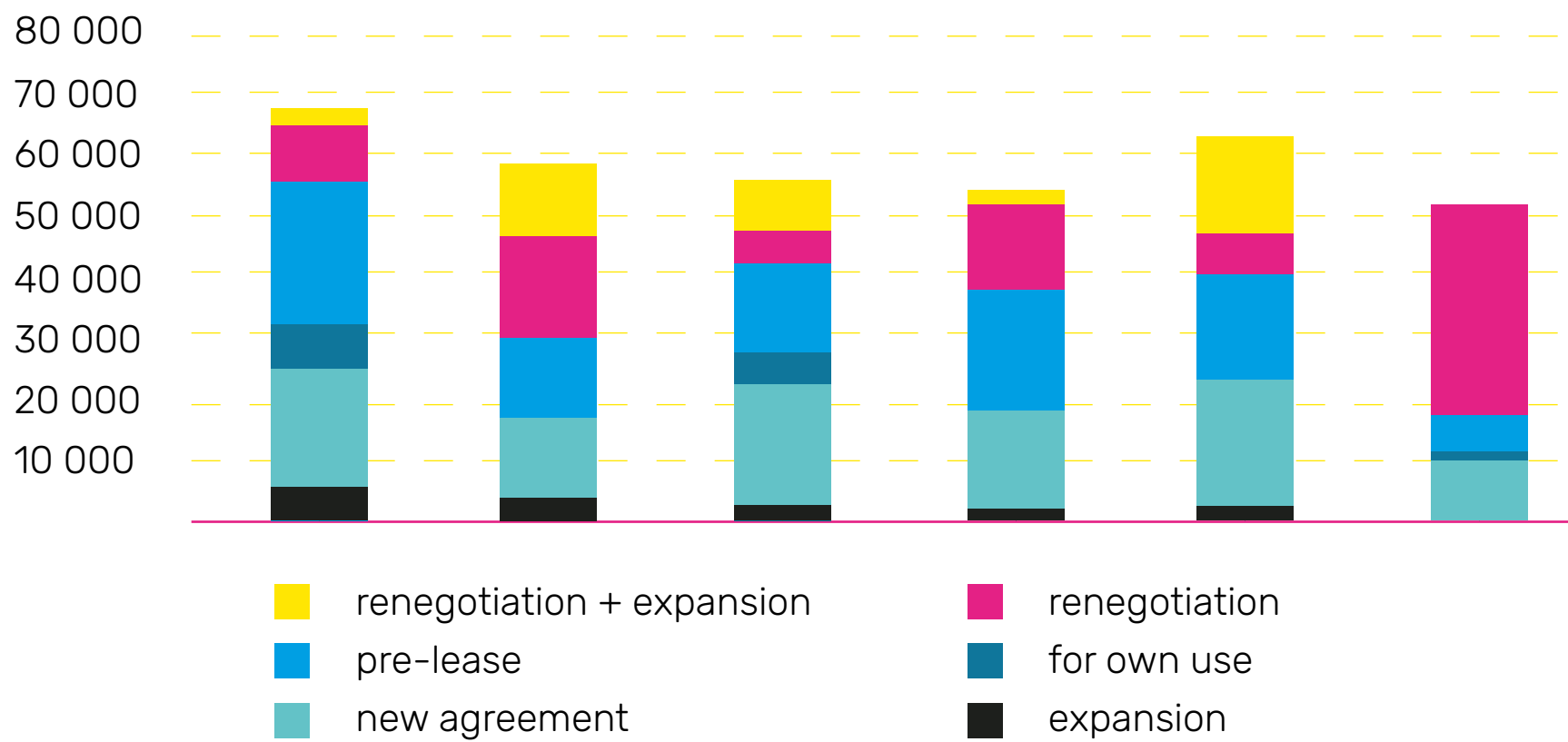
The critical change concerns the office function towards a place optimized for the organization's needs, focused on supporting integration and cooperation, in which desks are shared and away from assigned workstations.

Pilot study supplementing the qualitative part in the form of a questionnaire, number of respondents: 6

Office buildings in Lodz

The office landscape of Lodz has a positive effect on strengthening the city's image as an attractive work environment. Over the next two years, more modern office buildings will be commissioned, offering a high standard of finish and adequately fitting into the revitalized urban fabric of Lodz. The significant investments to be put into operation are Fuzja, Widzewska Manufaktura B (2023), Monopolis M2, and Widzewska Manufaktura A (2024). This year, the React project hit the market.

Offices showcase Lodz as an effective tool for promoting the city's investment attractiveness, especially in the eyes of foreign investors who expect the possibility of operating in well-located functional buildings. In addition, the standard of facilities is fundamental, as it provides attractive workplaces for employees and, at the same time, distinguishes the tenants' offer as employers who care about the comfort of work of people employed in their organizations.

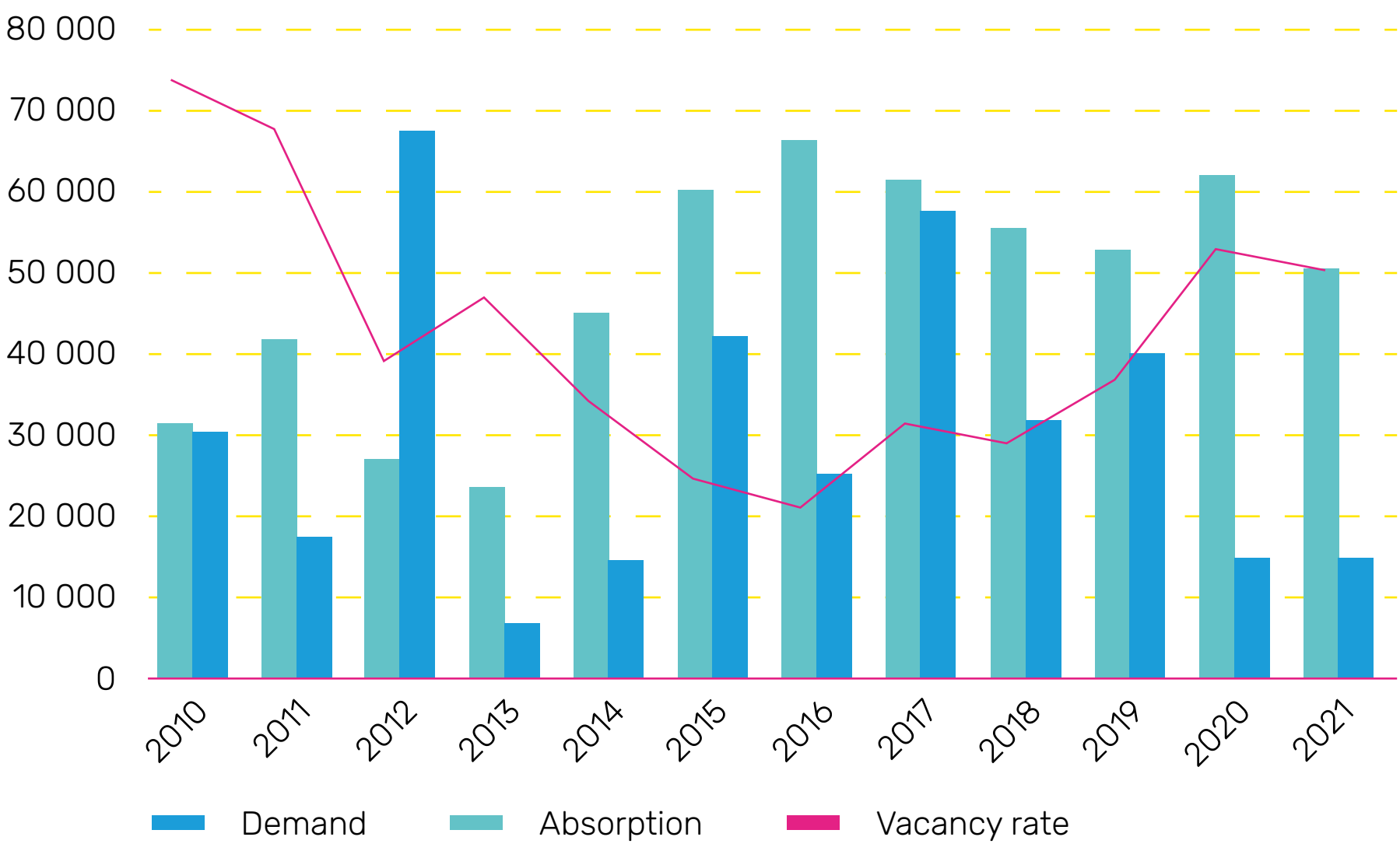
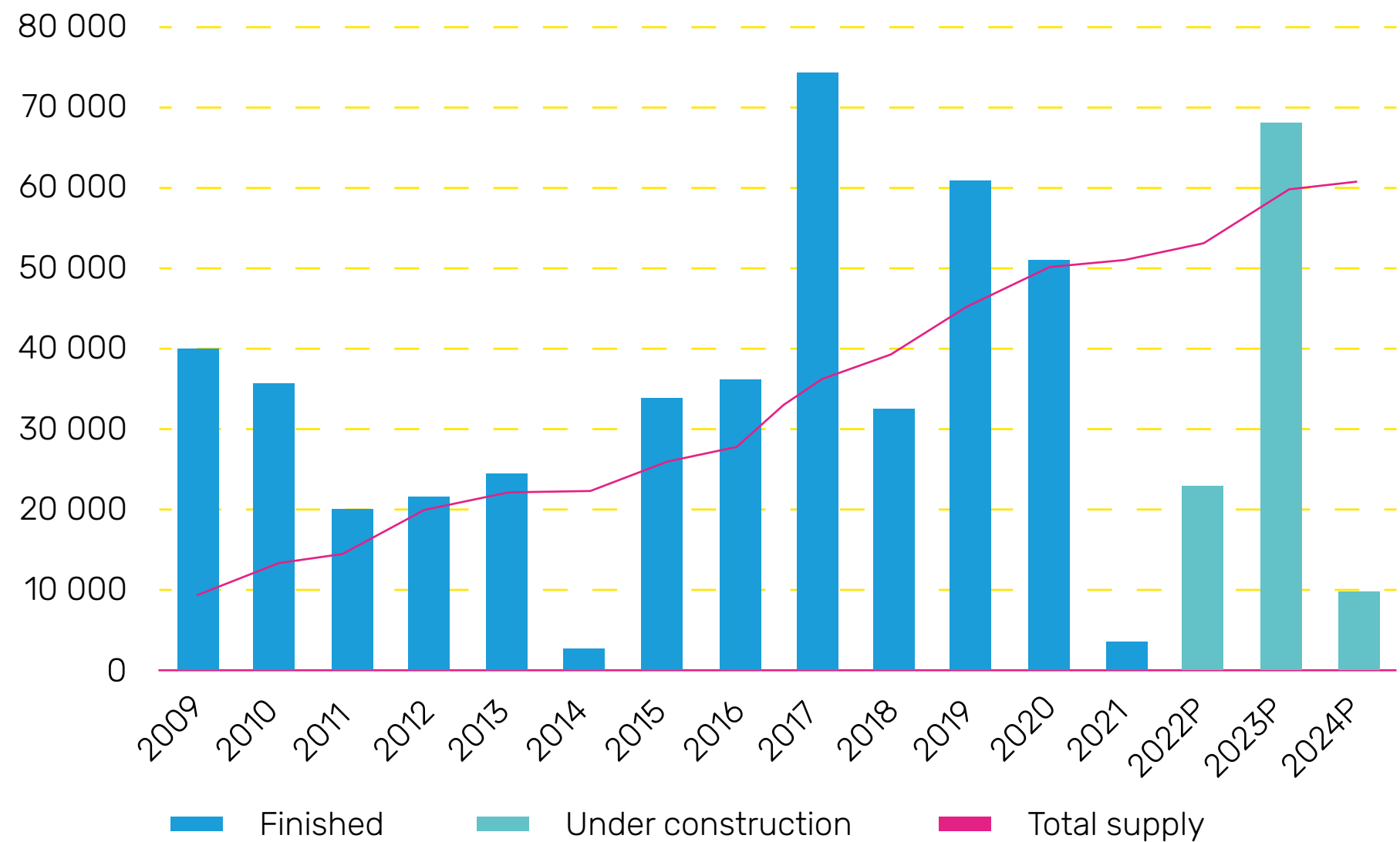


The potential of the Lodz office space market

The office market in Lodz has been growing regularly in recent years. The total supply of modern office space in the first quarter of 2022 amounted to 611,600 m², with over 77,900 m² under construction. At the same time (in the first quarter of this year), there were new lease agreements for 12,140 m² of office space in Lodz. As a result, the total demand in 2021 amounted to 50,700 m² and was mainly related to renegotiating existing lease agreements.

The last period in developing the office market in Lodz can be considered favorable for investors. In the previous five years, over 220,000 m² has been delivered. New office space and the annual demand did not fall below 50,000 m². Additionally, by the end of 2024, over 77,900 m² of new space is planned.

The evolution of the Lodz office space market



The potential of the city’s office space market in the opinion of the Lodz CBRE team

The stable development of the office market in Lodz makes the situation of investors and tenants comfortable. The availability of office space is sufficiently high. It allows for the development of both the development needs of current tenants operating in Lodz - especially in the IT industry, which is developing very dynamically in Lodz - and foreign business organizations that increasingly choose Lodz as a new location for conducting business activities.

Anna Trębicka

Director, Office Agency, CBRE

Availability and costs of office space in Lodz, including flexible space, make Lodz stand out on the map of large regional cities in Poland. Thus, the real estate market may be an element of the growing competitiveness of Lodz, including the growing investment attractiveness for foreign companies considering Lodz as a city for relocation of business processes in the future

Marta Pyziak

Associate Director, Office Agency, CBRE

Examples of locations offering a high standard of office space in Lodz are the buildings such as Ogrodowa Office and Hi Piotrkowska. Office tenants appreciate not only their architectural values but also the location, amenities related to access to public transport, and high construction standards in environmental protection. As a result, buildings have been awarded the BREEAM certificate at the Excellent level, thanks to which they are among the buildings with the highest sustainable construction standards.

Marta Zgłobicka

Associate Director, CBRE, Landlord Representation





Lodz IT ecosystem,
as seen by the
ICT Central Poland Cluster



Examples of ICT projects

Lodz universities associated with the ICT Central Poland Cluster - Lodz University of Technology, University of Lodz, and the Social Academy of Sciences have been running projects in various areas related to ICT for many years.

These are projects in teaching and those of a scientific and implementation nature. These grants are financed under the programs of domestic institutions (e.g., National Center for Research and Development, National Science Center, Information Processing Center), foreign programs, and retail orders.

In addition, many projects are carried out by consortia associated with leading domestic and foreign research units and enterprises. International projects are implemented mainly from the funds of the HORIZON 2020 program for units of the European Commission, incl.

- European Research Executive Agency (REA): *TOMOCON project - Intelligent tomographic sensors for advanced industrial process control,*
- European Innovation Council and SMEs Executive Agency (EISMEA): *MISEL project - Multispectral intelligent vision system with embedded low-power neural processing,*
- Directorate-General for Communications Networks, Content and Technology (CNECT): *Sound of Vision project - Natural senses of seeing through acoustics and haptics.*

There are several examples of international projects developing soft skills and shaping attitudes, implemented with the funds of the ERASMUS + program. To mention just a few:

- *Start IT: Development of Soft and Future Skills using Digital Entrepreneurship, Augmented Reality for Management Skills Development with real-based Cases,*
- *ACeSYRI: Advanced Center for Ph.D. Students and Young Researchers in Informatics,*
- *JEU: Jouons Ensemble pour l'Union Européenne - Citizen Games.*

Projects in the field of ICT carried out by scientists from Lodz universities support the development of various areas of social life, business, and public administration activities. Examples of such projects in the health care field are:

- *ASsIST* - a personalized intelligent telecare system,
- *MDB - MEDICAL DATA BANK* - a virtual platform for medical data and modern diagnostics.

The ICT Cluster also offers support for solutions intended for business. Including projects like:

- *The environment for the construction and operation of secure applications operating in the cloud based on intelligent anomaly detection in computing clusters and blockchain cryptographic techniques,*
- *Data Science in Practice,*
- *Improving the efficiency of logistics processes thanks to the development of universal methods of solving advanced methods of determining travel routes.*

In the field of public administration, including IT education at the secondary school level, our cluster supports the projects like:

- *Digital Commune,*
- *IT Championship Center, and*
- *Program the Future.*



Professor Paweł Strumiłło Ph.D., D.Sc.
Chairman of the ICT Central Poland Cluster

Members of ICT Central Poland



Commerzbank AG shares views about Lodz IT landscape

The IT industry in Lodz is constantly developing. It is reflected in the ever-growing number of employees of companies operating in Lodz. Success attracts people, and we see it in the number of new companies opening branches in our city.

Take Commerzbank as an example. Our company established a branch in Lodz in 2016. We made a decision based on the previous positive experience of operating in Lodz by a sister company from the Commerzbank Group. Knowledge of the local market and the availability of university graduates and office space were a sound basis for the further development of the branch in Lodz. During this time, we have grown from a team of a dozen people in 2016 to over 600 employees. It illustrates well the possibilities of developing the IT business in Lodz. This year, we are planning a further increase by a hundred and several dozen people, and we believe it will be possible.

The Covid-19 pandemic is changing the rules of the market. Remote work proves its potential. In this context, it isn't easy to talk about the advantages of the labor market in Lodz because, in practice, most IT companies operate on the domestic market, not the local one, related to the office address.

Marketing activities and the city's involvement in attracting new investors and companies impact their business decisions significantly. Undoubtedly, entering a new large employer into the market introduces turbulence (despite the city authorities' assurances that new companies are moving their operations from other locations along with their employees). Still, you have to learn to live with it. Our company has no problem with growing competition: we always operate in market conditions, and the competition strongly motivates us to implement improvements. Moreover, Lodz is a city conducive to start-ups. So if someone has a good idea for their business, they can take advantage of the competition program „An idea for a start-up.”

Pilot study supplementing the qualitative part in the form of a questionnaire, number of respondents: 6



In recent years, Lodz has significantly improved its image as a city friendly to residents, although there is still a long way to go to total happiness. The promotion of Lodz in the national arena could be better. Still, for it to be effective, the promotion must be accompanied by facts about the city's quality and attractiveness of life. There is still room for improvement and change here. This conclusion is confirmed by our observations regarding the recruitment and relocation of candidates to Lodz. While applicants from outside Poland are eager to relocate to Lodz, applicants living in Poland are less keen to relocate to Lodz, as for now.

Recruiting employees for IT companies is difficult regardless of where they operate. The shortage of specialists is omnipresent. Taking into account remote/hybrid work options used widely now, Lodz has lost its advantages related to lower business costs.

However, it is worth noting that Lodz gives employers an access to young talents. The „Youth in Lodz” program has been popular among young people and companies for many years. Every year, Lodz universities release a relatively large number of IT graduates. Summer internships and joint activities with universities as part of student internships and projects are a source of suitable candidates for work. Also, employers from the IT industry gathered in the ICT Central Poland Cluster have an impact on educating students at the IT faculties of the Lodz University of Technology, the University of Lodz, and the Social Academy of Sciences.

There is a need for more specialists in specific technologies or applications in the local market. Specifically, in our case, it concerns seasoned developers with a deep understanding of banking issues and applications. In such a situation, we use specialists from other regions (Warsaw, Wroclaw, Tricity), offering them remote work (teleworking under the current labor code). As a result, about 4% of our employees have this form of work.

Despite the changing conditions, Lodz remains a good location for the IT sector, offering a friendly business environment to companies and various opportunities for adepts to enter the professional path of development, as well as new challenges and further career development for experienced IT specialists.

Marek Gajowniczek

*Head of DTC Lodz,
Poland Commerzbank AG*



Artificial intelligence developmnet trends

Artificial intelligence (AI) technologies are, next to cloud technologies and the 5G telecommunications standard, the most critical drivers of the development of modern technology. According to the IEEE study - The impact of Tech in 2022 and beyond (conducted among leaders of technology companies) - AI is a top priority for developing global technology and the economy. Therefore, AI development trends can be considered, on the one hand, in the area of its applications in the economy, and on the other, in science and technology, where new and improved AI methods are developed.

In applications, it is worth pointing to the trend called „Augmented Intelligence,” which consists of increasing human work efficiency (augmented workforce). Techniques to support and accelerate the career of programmers and specialists testing software quality are dynamically developing.

Tools such as GitHub Copilot, now available in test mode, will significantly reduce software development time. AI applications in the area of cybersecurity radically reduce the risk of digital attacks through more effective and earlier threat identification. An equally strong trend is using NLP (Natural Language Processing) tools to significantly shorten the time of analyzing extensive legal acts. According to many automotive market analysts, 2022 may also be a turning point for autonomous cars. The leader of this market segment, Tesla, announced the launch of the „beta” version of the FSD (Full Self-Driving) car.

In AI's scientific and technological development, the so-called „transformers as deep learning models” deserve special attention. They are used, among others, in natural language processing. For example, the GPT-3 system (Generative Pre-trained Transformer 3) is „trained” on over 175 billion parameters. Impressive results are achieved by using transformers for image analysis, e.g., with the ViT network - Vision Transformer. An important development trend of AI is XAI (eXplainable Artificial Intelligence) techniques, in which the „black box” model is abandoned in teaching neural networks. In AI hardware solutions, attention is paid to the construction of giant specialized processors - such as in the TPU (Tensor Processing Unit) system or in the largest integrated circuit Cerebras Wafer Scale Engine, built of 850,000 elementary processors (cores). Thanks to parallel data processing, these solutions enormously speed up calculations and open up new areas of AI applications.

On the border of application and scientific development, there is a trend related to virtual reality, especially in the context of the Metaverse initiative, in which AI will play a key role. The AI will support GameDev processes - focused on the development of games and the creation of virtual worlds and new AI algorithms for the needs of self-learning games.

Mirek Sopek PhD
MakoLab Founder

MakoLab



Lodz University of Technology – digital competences for society

Lodz University of Technology focuses on the development of modern technologies which will foster our graduates' employability in the rapidly changing labour market not only in the Lodz region but also across the country and beyond. Our educational offer in IT encompasses nine study programmes (including three new!), built and conducted in close cooperation with the Central Poland ICT Cluster and foreign partners such as Harvard University, LMU Munich, Chalmers Goeteborg or NUS Singapore. Lodz University of Technology is a socially responsible university – we stimulate the growth of digital skills among children, youth, adults and seniors by engaging them in numerous educational initiatives undertaken by our academic staff.

Associate Professor
Andrzej Romanowski Ph.D., D.Sc.
Vice-Rector for Education, Lodz University of Technology



Studies in the IT field at Lodz University of Technology

Studies in Polish:

- Computer Science
- Applied Computer Science
- Information Technology
in Environmental Protection
- **NEW!** Artificial Intelligence
and Machine Learning

Studies in English [IFE]:

- Computer Science (ABET accredited)
- Information Technology (ABET accredited)
- Computer Science and Information Technology
- **NEW!** Modelling and Data Science
- **NEW!** Human-Computer Interaction

High quality of education confirmed by the PKA Education Excellence Certificate in the „Partner for Development” category, the KAUT quality mark, and the EUR-ACE Label.

TUL can also boast of two American ABET (Accreditation Board for Engineering and Technology) accredited programs

Postgraduate studies:

- Data Science
- Computer graphics and multimedia techniques
- Modern Java/Jakarta EE business applications



Events & initiatives

International competitions:

- MS Imagine Cup
- Cisco Switch Up Challenge
- ABB IT Challenge
- World Cup of Computer Implemented Inventions
- ACM student competitions

Regular events:

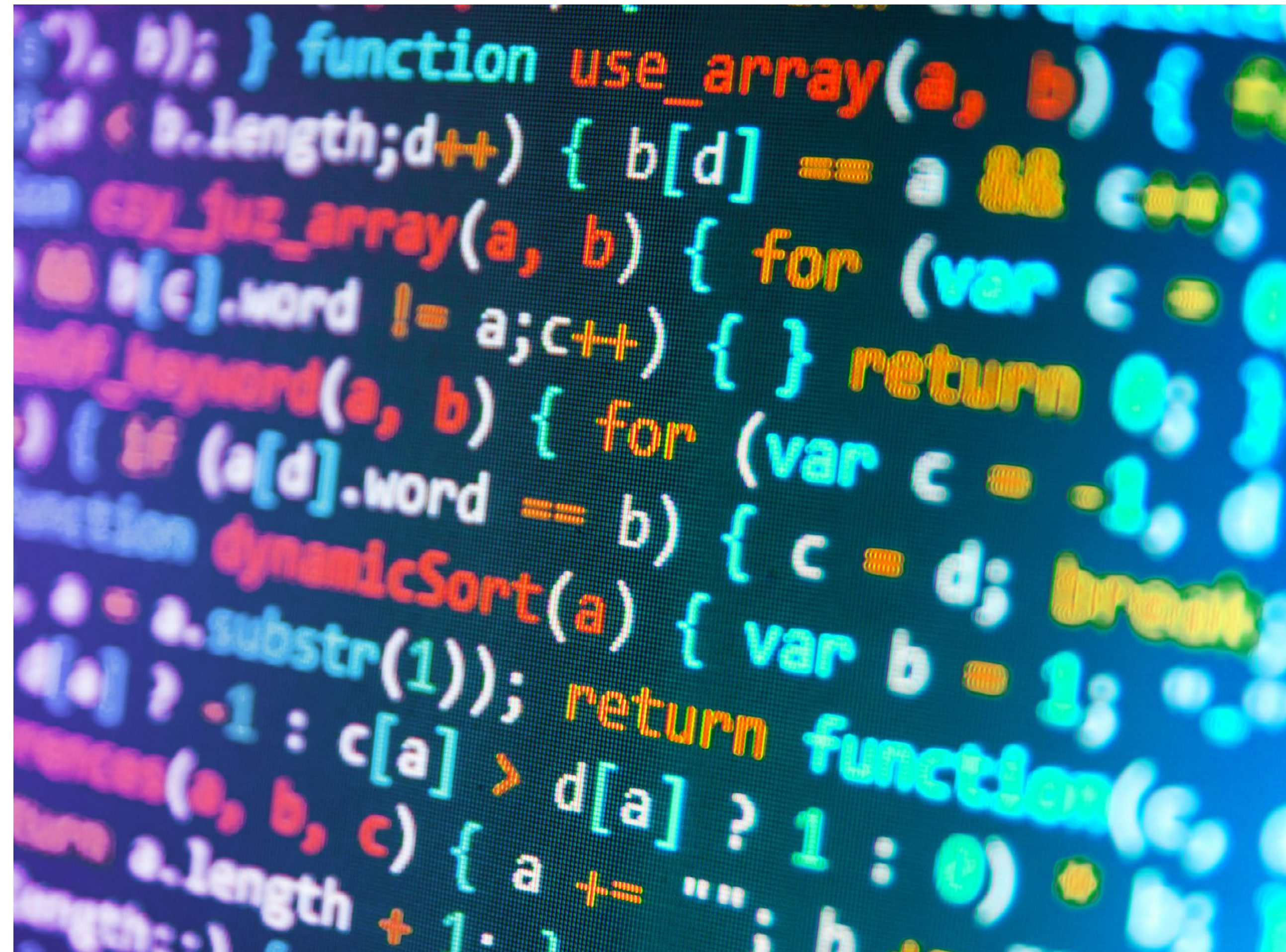
- Lodz IT Days
- IT Championship Center
- National Team Game Development Competition
- Scientific conferences (ACM, PTI)

Regional competitions:

- PIAP Safety Challenge
- Black See Science
- EkSoc StartUP
- GPT-3 Deep Learning Labs
- SAS Data Science hackathons

Student scientific clubs:

- SKN.NET
- Main
- UbiCOMP
- KINo Scientific clubs win projects and external funds (including grants from the Minister of „The Best of the Best! 4.0”). In addition, many outstanding IT students win awards and individual scholarships



ICT at the University of Lodz

The University of Lodz educates nearly 2,000 students in fields related to ICT, and each year 350 IT graduates leave our university. Our studies in the IT sector are primarily up-to-date with the situation in the labor market and created in close cooperation with business. The student can also count on the support of their interests at the University of Lodz because we know that the industry is changing dynamically, and genuine passion is the best tool to keep up to date. The University of Lodz has 12 different Faculties where you can find inspiration and people who help in development. **It is impossible to predict the following specializations in the computer and software industry today, so you have to do what we consistently promote - free your mind.**

Associate Professor
Agnieszka Kurczewska, PhD

*Vice-Rector for External Relations
University of Lodz*



Studies in the IT field at the University of Lodz

The field of **Computer Science** at the Faculty of Physics and Applied Computer Science, University of Lodz (WFILS) is a response to the constantly growing demand for specialists in the IT industry. Consequently, **learning outcomes are adjusted to the current needs of the labor market**. It is confirmed by the constantly growing number of candidates and graduates.

Students are offered modern specializations, such as **systems and mobile applications, algorithms and programming, databases and internet applications, or the administration of networks and information systems**. The choice of specializations and a wide range of optional subjects allow students to develop individual interests. For example, for 2nd-degree computer science students, the following paths are proposed:

- mobile systems.,
- artificial intelligence, or quantum computing.



At the beginning of 2020, the University of Lodz and seven strong business partners opened a new field of study – Business Process Automation. As a result, students of the Faculty of Management can acquire managerial competencies in connection with the knowledge of automation and robotization of business processes. Furthermore, it results from the employers' demand for RPA (**Robotic Process Automation**) specialists with well-developed digital competencies. Thanks to the combination of these qualifications, graduates will be able to understand modern digital technology and the possibilities of its application in business.

Practitioners conduct classes in **Business Process Automation** from companies dealing with automation and robotization: **Nordea, DXC, Digital Workforce, Digital Teammates, B/S/H, Fujitsu,** and **Mindbox**. Students finish their studies by creating a practical project for companies from the IT environment – student internships are closely related to the final project preparation. BPA is Poland's first field of study that comprehensively covers the subject of automation and robotization.

Cooperation with the business:

- Smart Technologies Center and the CAICR Center for Artificial Intelligence and Cyber communication Research, closely cooperate with ICT companies in the Lodz region. They conduct inspiring research, including in the international dimension, which contributes to deepening knowledge and strengthening cooperation between various research centers and the University of Lodz, enterprises, and private sector entities.

The rapidly progressing computerization and development of information technologies – observed for over a dozen years – enable the collection and processing of an increasing amount of data. However, some negative phenomena can also be observed, such as the formation of information noise due to excess information. In these conditions, extracting meaningful data from the point of view of optimizing economic and business processes, solving problems of state administration, or conducting scientific research is more challenging. To meet these challenges, **educating specialists with the skills to search, select and process information** is necessary. The answer to this labor market demand is studied in the field of **Data Analysis** offered by the Faculty of Mathematics and Computer Science at the University of Lodz.

#LODZfeelsIT

Finishing remarks

The Lodz IT ecosystem consists of committed professionals, open to challenges and looking for efficient solutions.

The basis for cooperation with our partners is authentic and direct relationships, co-creation, and mutual respect. In our daily activities, we focus on integrating the City with the economic environment and supporting the development of the Lodz IT ecosystem, meeting its ideas, needs, and expectations. Our partners and their development are our priority – for us, it is synonymous with city development.

Our Invest in Lodz team (part of the City of Lodz Office) is fully involved in each project. We approach every new strategy with the same passion and energy.

We invite you to join our Lodz IT ecosystem. In Lodz, ideas, energy, and passion bring fantastic results!

Adam Brzostowski

*Director, Business Development
and International Relations Bureau,
City of Lodz Office*



Acknowledgements



Adam Brzostowski
Director, Business Development and International Relations Bureau, City of Lodz Office
a.brzostowski@uml.lodz.pl



Renata Biadala
Project Manager, Business Development and International Relations Bureau, City of Lodz Office
r.biadala@uml.lodz.pl



Marta Sakowska
Manager New Business Initiatives Team, Business Development and International Relations Bureau, City of Lodz Office
marta.sakowska@uml.lodz.pl



Mariusz Lemparty
Branch Manager, Grafton Recruitment
mariusz.lemparty@grafton.pl



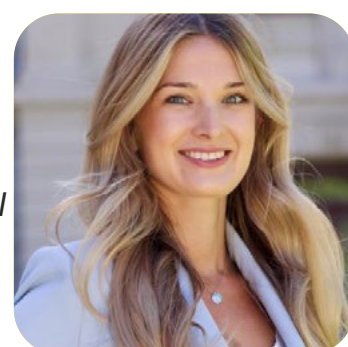
Joanna Ciężkowska
Brand Manager, Grafton Recruitment
joanna.ciezowska@gigroup.com



Anna Kęłowska-Staniszevska
Project Leader, Business Development and International Relations Bureau, City of Lodz Office
an.keblowska@uml.lodz.pl



Marta Hereżniak, Ph.D.
New Business Initiatives Team Business Development and International Relations Bureau
herezniak@uml.lodz.pl



Aleksandra Cebelińska
Development & Investor Relations Team Manager Business Development and International Relations Bureau
a.cebelinska@uml.lodz.pl



Micha Kramek
Data Expert, Grafton Recruitment
michal.kramek@grafton.pl



Agata Jemiola
Branch Manager, Grafton Recruitment
agata.jemiola@grafton.pl



Prof. Paweł Strumillo
Chairman, ICT Central Poland Cluster
pawel.strumillo@p.lodz.pl



Adam Owczarek
Manager ICT Central Poland Cluster
adam.owczarek@p.lodz.pl



Jan Banasikowski
Associate Director, Office Agency, CBRE
jan.banasikowski@cbre.com



Marta Pyziak
Associate Director, Office Agency, CBRE
marta.pyziak@cbre.com



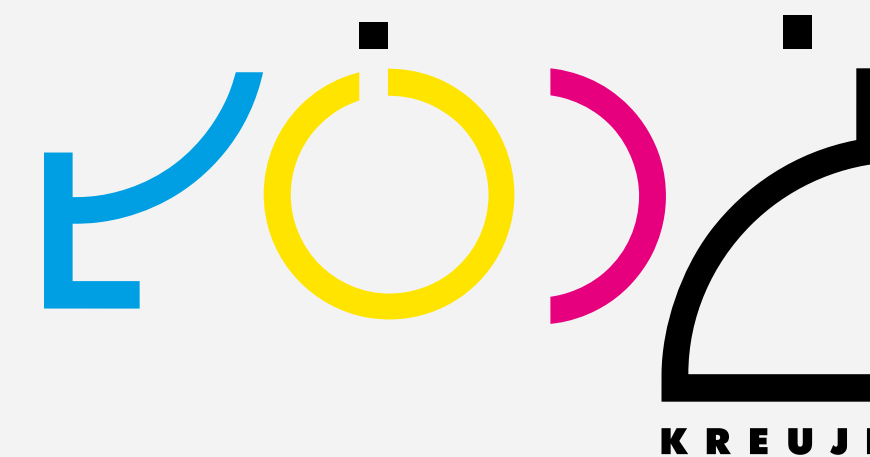
Agata Mróz
Senior Workplace Consultant, Workplace Strategy & Change Management, CBRE
agata.mroz@cbre.com



Marta Zglobicka
Associate Director, Landlord Representation, CBRE
marta.zglobicka@cbre.com



Anna Trębicka
Associate Director, Office Agency, CBRE
anna.trebicka@cbre.com



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